

ASYMPTOTIC ANALYSIS OF AN L-SHAPED JUNCTION OF TWO ELASTIC BEAMS

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UDC 517.956.8:517.958:539.3(5)

We propose a one-dimensional asymptotic model of an L-shaped junction of two thin two-dimensional elastic beams subject to boundary conditions of different type at external beam ends. The beams are described by standard systems of the Kirchhoff ordinary differential equations, whereas the transmission conditions on coincident (internal) nodes essentially depend on the type of boundary conditions on the external beam ends. The transmission conditions are obtained by analyzing the boundary layer near internal nodes. The asymptotics is justified with the help of a weighted anisotropic Korn inequality. Bibliography: 28 titles. Illustrations: 5 figures.

1 Introduction

1.1. Motivation

We consider the deformation problem for two thin two-dimensional beams joined at the right angle in the form of an elastic L-shaped junction. The asymptotic analysis of thin elastic bodies is rather complicated because of the behavior of plates and rods is different under bending and longitudinal deformation: it is easier to bend than to stretch. Therefore, the choice of the initial asymptotic ansatz for the displacement vector is of great importance (for example, in [1, 2], the improper equalization of orders of all the displacement components causes extra symmetry conditions on the shape and physical properties of thin anisotropic rods). At present, correct asymptotic ansätze for a single plane beam or a single three-dimensional rod are well known (cf., for example, [3]–[7]), which makes it possible to find simple and “almost explicit” formulas for

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effective moduli in one-dimensional models of beams and rods such as isotropic and anisotropic, straight and slightly curved, homogeneous and composite, laminated and possessing rapidly oscillating geometric and mechanical characteristics.

There are many unrelated results (cf. [8]–[13]) concerning junctions of thin beams and rods which are essentially depend on a particular shape and way of clamping. On the other hand, although a general scheme is known [14]–[16], this scheme is rather cumbersome and, consequently cannot be easily realized in particular situations. We mention the paper [17] (cf. also the monograph [18]) dealing with two beams with “external” clamped ends that are joined at a certain angle in an “internal” node (cf. Figure 1). This construction is considered by other method in Section 2. In this paper, this topic is introductory and is designated to explicate the asymptotic procedures described in Sections 3 and 4. The goal of this paper is to demonstrate how the asymptotic expansions of displacement and stress fields depend on whether the external beam ends are clamped. For this purpose we consider the simplest L-shaped junction. According to our goal, we pay the main attention to the description of asymptotic procedures, whereas the traditional justification of formal expansions is given at the end of the paper.

1.2. Junctions of elastic beams

Let $Q_{\pm}(h)$ be two (plus and minus) thin rectangles described in the local coordinates $(y_{\pm}, z_{\pm})$ as follows (cf. Figure 1): $Q_{\pm}(h) = \{(y_{\pm}, z_{\pm}) : |y_{\pm}| < h/2, -h/2 < z_{\pm} < l_{\pm} + h/2\}$.

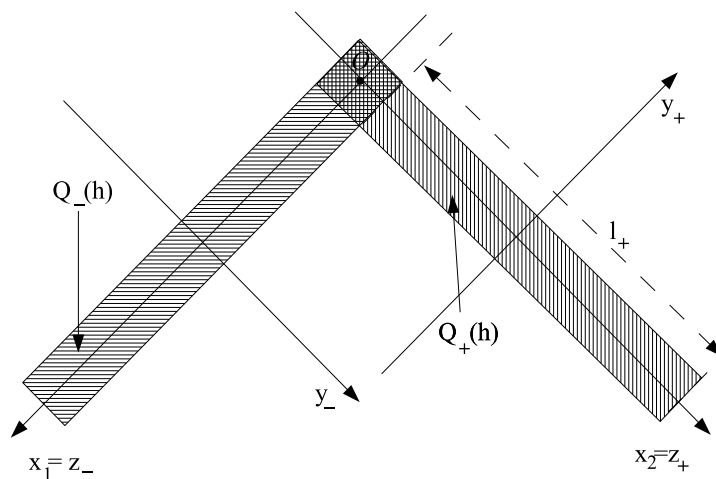


FIGURE 1. L-shaped junction and coordinate system.

The width $h > 0$ is a small positive parameter, and the scale is fixed, for example, by the condition $l_+ + l_- = 2$. We introduce the global Cartesian coordinates $x = (x_1, x_2)$ with center O and place the rectangles $Q_{\pm}(h)$ in such a way that the z_- and z_+ axes are directed along the x_1 - and x_2 - axes respectively, whereas the points $z_{\pm} = 0$ coincide with the origin O . We consider the junction $\mathcal{Q}(h)$ consisting of two beams $Q_{\pm}(h)$ and three nodes G_h^+ , G_h^- , and G_h^0 obtained by contracting h times the fixed sets G_1^0 and $G_1^{\pm}$ with piecewise smooth boundaries (cf. Figure 2) relative to the points $P_+ = (0, l_+)$, $P_- = (l_-, 0)$, and $P_0 = (0, 0)$ (the x -coordinates are indicated). To simplify the notation, we assume that $G_1^s \subset \mathbb{B}^s(1)$, where $\mathbb{B}^s(R)$ is the disk with radius R and center P_s and s takes the values 0 and $\pm$.

The “skeleton” $\mathfrak{S}$ of the junction

$$\mathcal{Q}(h) = Q_-(h) \cup Q_+(h) \cup G_h^+ \cup G_h^- \cup G_h^0 \quad (1.1)$$

consists of intervals $\Sigma_{\pm}$ of length $l_{\pm} + h$ and points $P_0, P_{\pm}$ (cf. Figure 3).

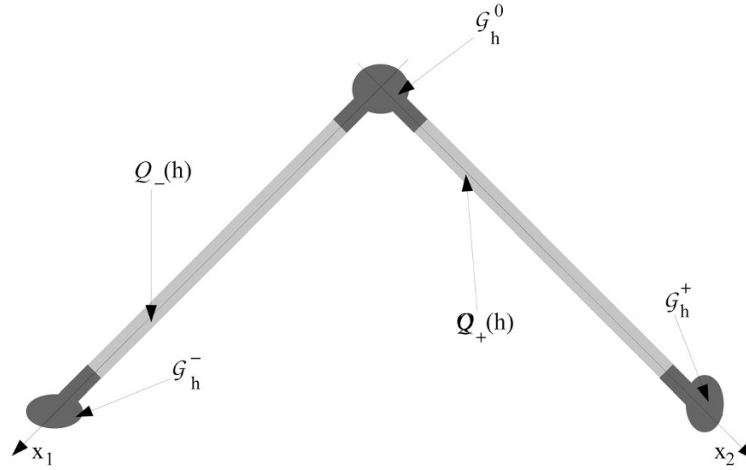


FIGURE 2. Junction $\mathcal{Q}(h)$. The nodes are toned.

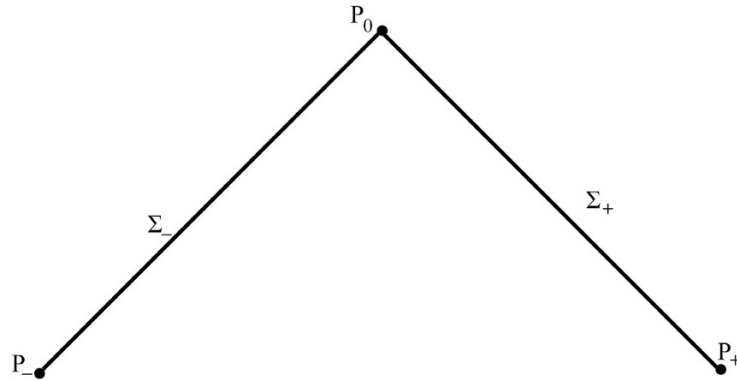


FIGURE 3. “Skeleton” $\mathfrak{S}$ of the system $\mathcal{Q}(h)$.

We extract shortened beams (rectangles) in $\mathcal{Q}(h)$

$$Q_+(h) = \{(x_1, x_2) : |x_1| < h/2, 2h < x_2 < l_+ - 2h\},$$

$$Q_-(h) = \{(x_1, x_2) : |x_2| < h/2, 2h < x_1 < l_- - 2h\},$$

and nodes (toned in Figure 2) of diameter $O(h)$

$$G_h^s = \mathbb{B}^s(3h) \cap (\mathcal{Q}(h) \setminus (Q_+(h) \cup Q_-(h))), \quad s = 0, \pm.$$

The extended nodes G_h^s are obtained by attaching to G_h^s thin rectangles located at the ends of the beams $Q_{\pm}(h)$. For the sake of convenience to describe asymptotic procedures and justify asymptotics we introduce the fragments $Q_{\pm}(h)$ and G_h^s of the junction $\mathcal{Q}(h)$ instead of $Q_{\pm}(h)$ and G_h^s .

In Section 4, we consider a special case of the junction $\mathcal{Q}(h)$: the set $\mathcal{Q}_0(h)$ consists of the beams $Q_{\pm}(h)$, triangular nodes

$$\begin{aligned}\mathcal{G}_h^+ &= \{(x_1, x_2) : |x_1| < h/2, |x_2 - l| < h/2, x_1 > x_2 - l\}, \\ \mathcal{G}_h^- &= \{(x_1, x_2) : |x_2| < h/2, |x_1 - l| < h/2, x_2 > x_1 - l\},\end{aligned}\quad (1.2)$$

and the square node

$$\mathcal{G}_h^0 = \{(x_1, x_2) : |x_1| < h/2, |x_2| < h/2\} \quad (1.3)$$

(cf. Figure 4). This geometry is required for setting a problem with periodicity conditions on the hypotenuses $\delta_h^{\pm}$ of the triangles $\mathcal{G}_h^{\pm}$.

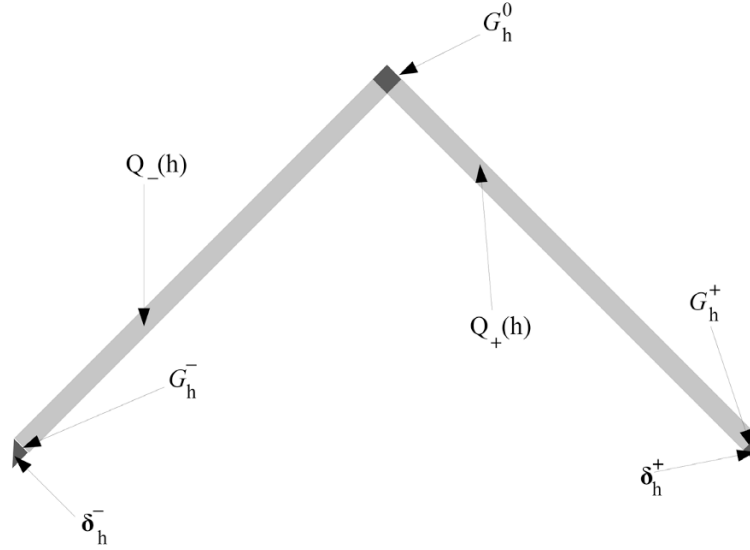


FIGURE 4. The set $\mathcal{Q}_0(h)$.

We consider the linear problem of the theory of elasticity for an anisotropic inhomogeneous body $\mathcal{Q}(h)$. We use the Voigt-Mandel notation [19]¹ for the defining relations, which is convenient for performing asymptotic analysis; moreover, the displacement vector $u = (u_1, u_2)^{\top}$ is interpreted as a column, i.e., $\top$ is the transposition symbol. We introduce the strain and stress columns

$$\begin{aligned}\varepsilon(u) &= (\varepsilon_{11}, \varepsilon_{22}, 2^{1/2}\varepsilon_{12})^{\top}, \\ \sigma(u) &= (\sigma_{11}, \sigma_{22}, 2^{1/2}\sigma_{12})^{\top}\end{aligned}\quad (1.4)$$

by expressing the Hooke law in the matrix form

$$\sigma(u) = A\varepsilon(u).$$

Here, $\varepsilon = (\varepsilon_{ij})$ denotes the strain tensor with Cartesian components

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right), \quad (1.5)$$

and the symmetric and positive definite 3×3 -matrix A is composed of the elasticity moduli of the anisotropic material. We assume that the entries of the matrix A depend only on the “fast”

¹ It is also associated with the name of S. G. Lekhnitsii (cf., for example, [20, 7]).

variable η on the beams $\mathcal{Q}_{\pm}(h)$. The factor $2^{1/2}$ is introduced to (1.4) in order to guarantee that the natural tensor norms coincide with the natural norms of the corresponding columns and the columns (1.4) are transformed by the orthogonal matrices under the orthogonal change of variables (cf. [7, Section 2.1]). By (1.4) and (1.5),

$$\varepsilon(u) = D(\nabla_x)^\top u,$$

where

$$D(\nabla_x)^\top = \begin{pmatrix} \partial/\partial x_1 & 0 & 2^{-1/2}\partial/\partial x_2 \\ 0 & \partial/\partial x_2 & 2^{-1/2}\partial/\partial x_1 \end{pmatrix}, \quad \nabla_x = \begin{pmatrix} \partial/\partial x_1 \\ \partial/\partial x_2 \end{pmatrix}.$$

In the Voigt-Mandel notation [20, 7, 19], the system of equations of the theory of elasticity takes the form

$$L(h, x, \nabla_x)u(h, x) := D(-\nabla_x)^\top A(x)D(\nabla_x)u(h, x) = f(h, x), \quad x \in \mathcal{Q}(h), \quad (1.6)$$

where $f = (f_1, f_2)^\top$ are the volume forces. The conditions on components of the vector-valued function f and entries of the matrix A will be given below as needed.

1.3. Junction of beams with clamped ends

In Section 2, we consider the junction (1.1) with fixed “external” nodes $\mathcal{G}_h^\pm$. In this case, we can use asymptotic ansätze for single beams with a slightly modified procedure of constructing asymptotics near the “internal” node $\mathcal{G}_h^0$ (cf. [17]). The resulting one-dimensional model of junction consists of two systems of ordinary differential equations of the second and fourth order for vectors of averaged longitudinal and transverse displacements

$$w^\pm(z_\pm) = (w_1^\pm(z_\pm), w_2^\pm(z_\pm))^\top \quad (1.7)$$

entering in the primary asymptotic ansatz on the beams $\mathcal{Q}_\pm(h)$

$$u(h, x)|_{\mathcal{Q}_\pm(h)} \sim h^{-2}w_2^\pm(z_\pm)e_2^\pm + h^{-1}(w_1^\pm(z_\pm) - h^{-1}y^\pm\partial_{z_\pm}w_2^\pm(z_\pm))e_1^\pm. \quad (1.8)$$

Hereinafter, $e_1^\pm$ are the unit vectors of the $z_\pm$ -axes and $e_2^\pm$ are the unit vectors of the $y_\pm$ -axes. At the points $P_\pm \in \mathfrak{S}$ representing the node $\mathcal{G}_h^\pm$, we impose the natural Dirichlet conditions, usual in the theory of beams,

$$w_1^\pm(l_\pm) = 0, \quad w_2^\pm(l_\pm) = 0, \quad \partial_{z_\pm}w_2^\pm(l_\pm) = 0. \quad (1.9)$$

At the point $P_0 \in \mathfrak{S}$ with coordinates $z_\pm = 0$, which represents the node $\mathcal{G}_h^0$, we impose six transmission conditions.² We describe these conditions which admit a simple physical interpretation.

First of all, the transverse displacements vanish:

$$w_2^\pm(0) = 0, \quad (1.10)$$

² Transmission conditions are usually understood as relations between the unknowns and their derivatives at the common point $z_\pm = 0$ of two coincident intervals. However, in the one-dimensional problem of the theory of elasticity under consideration, the transmission conditions also include the boundary conditions at disconnected points $z_+ = l_+$ and $z_- = l_-$.

whereas there are no restrictions on the longitudinal displacements $w_1^\pm(0)$. These conclusions agree with the ansatz (1.8): since the vectors e_2^+ and e_2^- are mutually perpendicular, the coincidence of the expressions (1.8) at the point P_0 with accuracy $O(h^{-1})$ requires the validity of (1.10). On the other hand, it is not necessary to connect somehow $w_1^+(0)$ and $w_1^-(0)$ since the asymptotic conglomerate ³

$$h^{-1}W_2^\pm(z_\pm)e_2^\pm + h^0(W_1^\pm(z_\pm) - h^{-1}y_\pm\partial_{z_\pm}W_2^\pm(z_\pm))e_1^\pm, \quad (1.11)$$

similar to the extracted one on the right-hand side of (1.8) and the next in the total asymptotic expansion of the solution $u(h; x)$, can be subject to the conditions

$$W_2^+(0) = w_1^-(0), \quad W_2^-(0) = w_1^+(0),$$

and thereby we can improve the adjustment of the ansätze on the coincident beam ends.

To treat the last terms in (1.8), we apply the Taylor formula. By (1.10), we get

$$u(h, x)|_{Q_\pm(h)} \sim h^{-1}w_1^\pm(0)e_2^\pm + h^{-2}(z_\pm e_2^\pm - y_\pm e_1^\pm \partial_{z_\pm} w_2^\pm(0)).$$

We can ignore the first term on the right-hand side (cf. comments to formula (1.10)). The factor at h^{-2} represents a rotation about the point P_0 , and its continuity under passage from one beam to another is equivalent to the transmission condition

$$\partial_{z_+} w_2^+(0) = \partial_{z_-} w_2^-(0). \quad (1.12)$$

The three transmission conditions (1.10), (1.12), and the six (three conditions on each external end of one-dimensional beams) Dirichlet conditions (1.9) are called *stable* in the terminology [21] since they are inherited by the function space in the variational setting of the limit one-dimensional problem. The remaining three transmission conditions, called *intrinsic*, appear in the Green formula as the dual of the stable conditions (cf. [21, Chapter 2, Section 2 and 9, Remark 2.9.5]). This method for obtaining transmission conditions is used in Section 2. The intrinsic conditions are interpreted from the point of view of mechanics as follows: at the node P_0 , the bending moment is continuous and both averaged longitudinal forces vanish (cf. (2.33)). We emphasize that the general results of [21] provide the solvability of the above limit problem and the smoothness improvement for solutions to this problem (Proposition 2.1).

There is a general approach to the definition of boundary and transmission conditions at endpoints of one-dimensional images of thin bodies (cf. [22] and [23, Chapter 16]) based on the decay property of terms of the boundary layer appearing, in the case under consideration, in neighborhoods of the external $\mathcal{G}_h^\pm$ and internal $\mathcal{G}_h^0$ nodes respectively. This approach is used in Section 3 to obtain the transmission conditions for junction of beams with free ends.

1.4. Junction of beams with free ends

By the results of [14, 16], the fact that the external nodes $\mathcal{G}_h^\pm$ are not fixed crucially changes even the leading term of the asymptotic ansatz (1.8) which takes the form

$$u(h, x)|_{Q_\pm(h)} \sim h^{-2}(w_2^\pm(z_\pm)e_2^\pm + a_1^\pm e_1^\pm) + h^{-1}(w_1^\pm(z_\pm) - h^{-1}y^\pm \partial_{z_\pm} w_2^\pm(z_\pm))e_1^\pm + \dots \quad (1.13)$$

³ i.e., a group of asymptotic terms of the same structure that are repeated from step to step of the procedure for constructing the total expansion of the solution.

(cf. Subsection 3.2 and formula (1.8)). The appearance of the constant terms $h^{-2}a_1^\pm e_1^\pm$ completely agrees with the fact that a solution to the problem about the body $\mathcal{Q}(h)$ with free boundary is found up to a summand (a rigid displacement), but the above terms affect the whole procedure of constructing asymptotics.

First, instead of the Dirichlet conditions (1.9) on the external beam ends $z_\pm = l_\pm$, we impose the Neumann conditions (3.33)–(3.35) which means that the bending moment, together with the transverse and cutting forces, vanishes. Second, because the ansatz (1.13) contains free constants $a_1^\pm$ which can be taken equal to $w_2^\mp(0)$, the boundary condition (1.10) becomes unnecessary. Since the rotation continuity condition as a unique stable transmission condition is preserved, it remains to find five intrinsic transmission conditions. The three ones are the same as in the case of the rigidly fixed nodes $\mathcal{G}_h^\pm$, i.e., in the case where the moment is continuous and the longitudinal forces vanish (cf. (3.25) and (3.26)). Taking into account that the condition (1.10) is removed and based on the above-mentioned scheme [21, Chapter 2, Section 9], it is easy to predict that the remaining two conditions are related to the averaged cutting forces on the internal beam ends. However, these boundary conditions turn out to be inhomogeneous. Therefore, to define the right-hand sides, we need once more modify the asymptotic ansatz by a specific lower order asymptotic conglomerate (1.11). The technical aspects are discussed in Subsection 3.2. Here, we explain the mechanical interpretation. Since the problem about the body $\mathcal{Q}(h)$ with free boundary is solvable only if three orthogonality conditions annihilating the principal vector and moment of external loads are satisfied, there are also three computability conditions in the one-dimensional limit problem. At the same time, computing the principal vector, we integrate over the union of two mutually perpendicular beams. Hence the sums of aggregate transverse loads $F_2^\pm e_2^\pm$ on one beam and the aggregate longitudinal loads $F_1^\mp e_1^\mp$ on the other beam vanish.

The vector $F_2^\pm e_2^\pm$ participates in the definition of bending $h^{-2}w_2^\pm(z_\pm)e_2^\pm$ of the beam $\mathcal{Q}_\pm(h)$, but the order h^0 of the vector $F_1^\mp e_1^\mp$ is too small to affect the leading term $h^{-1}w_2^\mp(z_\mp)e_1^\mp$ of the longitudinal displacement of $\mathcal{Q}_\mp(h)$. Therefore, we have to consider the lower order asymptotic conglomerate (1.11) and the problem for $W_1^\mp(z_\mp)$ with the right-hand side $F_1^\mp$. The displacement $W_1^\mp e_1^\mp$ generates the longitudinal force $F_1^\mp e_1^\mp$ at the end $z_\mp = 0$ of the beam $\mathcal{Q}_\pm(h)$, which can be compensated only by the transverse force at the end $z_\mp = 0$ of the beam $\mathcal{Q}_\mp(h)$, i.e., the forces $F_1^\mp$ appear on the right-hand side of the boundary condition (3.32). This fact is obvious in mechanics and is obtained, as in Subsection 3.4, from the decay property of the boundary layer. Furthermore, the limit problem for the initial terms (1.7) of the ansatz (1.8) becomes solvable: the computability conditions in the one-dimensional problems follow from similar conditions in the initial two-dimensional problem. Moreover, neither the components (1.11) nor the boundary layer enter the leading term of the justified asymptotics of elastic fields in the junction $\mathcal{Q}(h)$ (cf. Theorem 5.1), but the procedure of obtaining estimates for the asymptotic remainders is essentially based on the above constructions.

1.5. Junction of beams with periodicity condition

In Section 4, we consider a special junction with periodicity conditions on the faces $\delta_h^\pm$ of the construction $\mathcal{Q}(h)$ (cf. Figure 4). In other words, we study deformations of an infinite corrugated thin elastic strip under the action of periodic loads. Comparing with Section 3, the algorithm for constructing asymptotics is slightly modified: the rotation of $\mathcal{Q}(h)$ as a rigid body is not a periodic vector-valued function, i.e., only two computability conditions are left: the components

of the external load vector vanish. At the same time, the particularity of the problem with periodicity conditions on $\delta_h^\pm$ causes modifications of the boundary conditions in the limit one-dimensional problem: the functions $w_2^\pm$ become periodic and $w_1^\pm$ become antiperiodic:

$$w_2^\pm(0) = w_2^\pm(l_\pm), \quad w_1^\pm(0) = -w_1^\pm(l_\pm). \quad (1.14)$$

The condition (1.12) expressing the continuity of rotation about the point $z_\pm = 0$ is preserved, and, in addition, a similar condition relative to the point $z_\pm = l_\pm$ (external beam ends are identified by periodicity) is imposed:

$$\partial_{z_+} w_2^+(l_+) = \partial_{z_-} w_2^-(l_-). \quad (1.15)$$

The intrinsic conditions, dual to the stable conditions (1.14) and (1.12), (1.15) in view of the Green formula, mean the periodicity of transverse force, the antiperiodicity of longitudinal force, and the continuity of moment at the points $z_\pm = 0$ and $z_\pm = l_\pm$. As in Section 3, in order to describe completely the intrinsic transmission conditions, we need to construct the lower order asymptotic conglomerate (1.11), where the antiperiodic functions $W^\pm$ on the beams $\mathcal{Q}_\mp(h)$ are uniquely found and transfer the longitudinal forces $F_1^\mp$ to the jump conditions on transverse forces at the endpoints $z_\pm = 0$ and $z_\pm = l_\pm$ of the beams $\mathcal{Q}_\pm(h)$, which thereby provides the existence of periodic functions $w_2^\pm$.

The above-listed nuances of asymptotic analysis of the periodic problem do not affect the algorithm described in Section 3. Therefore, we restrict ourselves to a brief summary in Section 4.

The justification procedure for the formal asymptotics constructed in Sections 2–4 plays a secondary role in our paper and is presented in Section 5 only for the problem discussed in Section 3. The matter is that the case of fixed nodes $\mathcal{G}_h^\pm$ is significantly simpler (in fact, the same as in the case of a single beam with clamped ends) and only slight modifications are required in the periodic case. Finally, a weighted anisotropic Korn inequality for an L-shaped junction of beams can be easily obtained from the results of [14] (cf. Subsection 5.1).

2 Junction of Beams with Clamped Ends

2.1. Statement of the problem

On the boundaries of external nodes, we extract small arcs $\Gamma_h^\pm$ by contracting h times fixed arcs $\Gamma_1^\pm \subset \partial\mathcal{G}_1^\pm \setminus \overline{\mathcal{Q}^\pm}(1)$ and impose the rigid clamping condition

$$u(h, x) = 0, \quad x \in \Gamma_h^\pm. \quad (2.1)$$

On the remaining part of the boundary $\gamma(h) := \partial\mathcal{Q}(h) \setminus (\overline{\Gamma_h^+} \cup \overline{\Gamma_h^-})$, which is free from external loads, we impose the homogeneous Neumann boundary condition

$$B(h, x, \nabla_x)u(h, x) := D(\nu(x))^\top A(h, x)D(\nabla_x)u(x, h) = 0, \quad x \in \gamma(h). \quad (2.2)$$

Hereinafter, ν is the outward normal unit vector defined almost everywhere on the Lipschitz surface $\partial\mathcal{Q}(h)$.

By a *weak solution* to the problem (1.6), (2.1), (2.2) in the Sobolev space $H^1(\mathcal{Q}(h))^2$ (the superscript “2” indicates the number of components) we mean an element u of the subspace

$$\mathring{H}^1(\mathcal{Q}(h), \Gamma_h)^2 = \{u \in H^1(\mathcal{Q}(h))^2 : u = 0 \text{ on } \Gamma_h := \Gamma_h^+ \cup \Gamma_h^-\}$$

satisfying the integral identity [24]

$$\mathcal{E}(u, v; \mathcal{Q}(h)) = (f, v)_{\mathcal{Q}(h)} \quad (2.3)$$

for any test vector-valued function $v \in \mathring{H}^1(\mathcal{Q}(h), \Gamma_h)^2$. Here, $(\cdot, \cdot)_{\Xi}$ is the inner product in the space $L_2(\Xi)$ (scalar or vector) and $\frac{1}{2}\mathcal{E}(u, u; \mathcal{Q}(h))$ is the elastic energy functional

$$\mathcal{E}(u, v; \Xi) = (AD(\nabla_x)^\top u, D(\nabla_x)v)_{\Xi}. \quad (2.4)$$

The following assertion concerning the solvability of the problem (1.6), (2.1), (2.2) is known.

Lemma 2.1. *For $f \in L_2(\mathcal{Q}(h))^2$ there exists a unique weak solution $u \in \mathring{H}^1(\mathcal{Q}(h))^2$ to the problem (2.3) and the following estimate holds:*

$$\|u; H^1(\mathcal{Q}(h))\| \leq c(h)\|f; L_2(\mathcal{Q}(h))\|. \quad (2.5)$$

The factor $c(h)$ on the right-hand side of (2.5) depends on the small parameter h , and the first step of the justification procedure for the asymptotics in Section 5 is to clarify this dependence by using a suitable weighted anisotropic Korn inequality [14].

2.2. Asymptotic ansätze on beams. Higher order terms

Assume that $\eta = (\eta_1, \eta_2) = (h^{-1}x_1, h^{-1}x_2)$ denote “stretched” coordinates and e_1 and e_2 are the unit vectors of the x_1 - and x_2 -axes. As usual (cf., for example, [3] and [4, 5, 7]), the asymptotic representations of displacement fields on thin elastic beams with clamped ends are taken as follows:

$$u(h, x) \sim h^{-2}U^{-2}(z) + h^{-1}U^{-1}(z, \eta) + h^0U^0(z, \eta) + h^1U^1(z, \eta) + \dots \quad (2.6)$$

On each beam $\mathcal{Q}_{\pm}(h)$, we set

$$U^{-2}(z) = w_2^{\pm}(z)e_2^{\pm}, \quad U^{-1}(z, \eta) = \left(w_1^{\pm}(z) - \eta \frac{\partial w_2^{\pm}}{\partial z}(z) \right) e_1^{\pm}. \quad (2.7)$$

Here, $z = z_- = x_1$, $\eta = \eta_- = \eta_2$ on $\mathcal{Q}_-(h)$ and $z = z_+ = x_2$, $\eta = \eta_+ = \eta_1$ on $\mathcal{Q}_+(h)$; moreover, $e_1^- = e_1$, $e_2^- = e_2$ and $e_1^+ = e_2$, $e_2^+ = -e_1$. The ansatz (2.6), together with the higher order terms (2.7) which were already mentioned in Subsection 1.3, reflect the following obvious fact: the beam $\mathcal{Q}_{\pm}(h)$ is deformed in the transverse direction $e_2^{\pm}$ easier than in the longitudinal direction $e_1^{\pm}$.

For functions $x \mapsto \Phi(z, \eta)$ we have the following obvious equality:

$$D(\nabla_x)\Phi(z, \eta) = (h^{-1}D_{\eta}\Phi(z, \eta) + D_z\Phi(z, \eta))|_{\eta=h^{-1}x}, \quad (2.8)$$

where $\partial_{\eta} = \partial/\partial\eta$, $\partial_z = \partial/\partial z$ and $D_{\eta} = D(e_2^{\pm}\partial_{\eta})$, $D_z = D(e_1^{\pm}\partial_z)$. Therefore, the operator L in (1.6) in $\mathcal{Q}_{\pm}(h)$ and the operator B in (2.2) on

$$\Sigma_{\pm}^q(h) = \{(\eta, z) : \eta = (-1)^q/2, \ h < z < l_{\pm} - h\}, \quad q = 1, 2,$$

admit the following expansions:

$$\begin{aligned} L(h, x, \nabla_x) &= h^{-2}L^0(x, \partial_{\eta}) + h^{-1}L^1(x, \partial_z, \partial_{\eta}) + h^0L^2(x, \partial_z), \\ B(h, x, \nabla_x) &= h^{-1}B^{0,q}(x, \partial_{\eta}) + h^0B^{1,q}(x, \partial_z), \end{aligned} \quad (2.9)$$

where

$$\begin{aligned} L^0 &= -D_\eta^\top AD_\eta, \quad L^1 = -D_\eta^\top AD_z - D_z^\top AD_\eta, \quad L^2 = -D_z^\top AD_z, \\ B^{0,q} &= (-1)^q D(e_2^\pm)^\top AD_\eta, \quad B^{1,q} = (-1)^q D(e_2^\pm)^\top AD_z, \quad q = 1, 2. \end{aligned}$$

We assume that the rigidity matrix A smoothly depends on the variable η . However, it is not difficult to consider the piecewise-smooth case (reinforced and laminated beams); cf. [7, Chapter 4].

Let $(h, x) \mapsto f(h, x)$ be a vector-valued smooth function of the parameter $h \in [0, h_0]$ and the point x in a fixed neighborhood of the junction $\mathcal{Q}(h)$. Consequently, on the beams $\mathcal{Q}_\pm(h)$,

$$f(h, x) = f^{0,\pm}(z) + \dots, \quad f^{0,\pm}(z) = f(0, 0, z) \quad (2.10)$$

where the dots denote a small remainder which is estimated in Subsection 5.3.

We substitute the expansions (2.6), (2.9), and (2.10) into Equation (1.6) and the boundary conditions (2.2) and collect coefficients at the same degrees of h . As a result, we obtain the recurrent sequence of one-dimensional problems on the cross-sections of the beams

$$L^0 U^k = F^k := -L^1 U^{k-1} - L^2 U^{k-2} + \delta_{k,2} f^{0,\pm}, \quad \eta \in \left(-\frac{1}{2}, \frac{1}{2}\right), \quad (2.11)$$

$$B^{0,q} U^k = G^{k,q} := -B^{1,q} U^{k-1}, \quad \eta = (-1)^q \frac{1}{2}, \quad q = 1, 2; \quad (2.12)$$

where $k = -2, -1, 0, 1$ and $U^p = 0$ for $p < -2$. Each of these problems is solvable if the following two conditions are satisfied:

$$\int_{-1/2}^{1/2} F_i^k(z, \eta) d\eta + \sum_{q=0,1} G_i^{k,q}(z) = 0, \quad i = 1, 2, \quad (2.13)$$

and the solution U^k is unique provided that

$$\int_{-1/2}^{1/2} U_i^k(z, \eta) d\eta = 0, \quad i = 1, 2. \quad (2.14)$$

The vectors (2.7) satisfy Equations (2.11), (2.12) for $k = -2$ and $k = -1$. Furthermore,

$$D_z U^{-1}(z, \eta) = \mathcal{Y}^\pm(\eta) \mathcal{D}(\partial_z) w^\pm(z); \quad (2.15)$$

moreover, $w^\pm = (w_1^\pm, w_2^\pm)$ (cf. (1.7)) and

$$\mathcal{Y}^-(\eta) = \begin{pmatrix} 1 & -\sqrt{2}\eta \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathcal{Y}^+(\eta) = \begin{pmatrix} 0 & 0 \\ 1 & -\sqrt{2}\eta \\ 0 & 0 \end{pmatrix}, \quad \mathcal{D}(\partial_z) = \begin{pmatrix} \partial_z & 0 \\ 0 & \frac{1}{\sqrt{2}} \partial_z^2 \end{pmatrix}. \quad (2.16)$$

By (2.11), (2.12), and (2.7),

$$F^0 = D_\eta^\top AD_z U^{-1}, \quad G^{0,q} = (-1)^q D(e_2^\pm)^\top AD_z U^{-1}.$$

Thus, from (2.15) and (2.16) it follows that

$$U^0(z, \eta) = \mathcal{X}^\pm(\eta, z) \mathcal{D}(\partial_z) w^\pm(z), \quad (2.17)$$

where $\mathcal{X}^\pm$ are (2×2) -matrices found from the problems

$$\begin{aligned} -D_\eta^\top A D_\eta \mathcal{X}^\pm &= D_\eta^\top A \mathcal{Y}^\pm, \quad \eta \in \left(-\frac{1}{2}, \frac{1}{2}\right), \\ (-1)^q D(e_2^\pm)^\top A D_\eta \mathcal{X}^\pm &= (-1)^{q+1} D(e_2^\pm)^\top A \mathcal{Y}^\pm, \quad \eta = (-1)^q \frac{1}{2}, \quad q = 1, 2. \end{aligned} \quad (2.18)$$

It is obvious that the compatibility conditions in these problems are satisfied. We assume that this matrix satisfies the orthogonality conditions (2.14), i.e.,

$$\int_{-1/2}^{1/2} \mathcal{X}^\pm(\eta) d\eta = 0. \quad (2.19)$$

2.3. Asymptotic ansätze on beams. The resulting system

For $k = 1$ we find

$$\begin{aligned} F^1 &= D_\eta^\top A D_z U^0 + D_z^\top A (D_\eta U^0 + D_z U^{-1}), \\ G^{1,q} &= (-1)^q D(e_2^\pm)^\top A D_z U^0. \end{aligned}$$

We verify the second ($i = 2$) compatibility condition in (2.13), namely,

$$\begin{aligned} (e_2^\pm)^\top \left(\int_{-1/2}^{1/2} F^1 d\eta + \sum_{q=0,1} G^{1,q} \right) &= (e_2^\pm)^\top \int_{-1/2}^{1/2} D_z^\top A (D_\eta U^0 + D_z U^{-1}) d\eta \\ &= \frac{\partial}{\partial z} \int_{-1/2}^{1/2} (D_\eta(\eta) e_1^\pm)^\top A (D_\eta U^0 + D_z U^{-1}) d\eta \\ &= -(e_1^\pm)^\top \frac{\partial}{\partial z} \left(\int_{-1/2}^{1/2} \eta D_\eta^\top A (D_\eta U^0 + D_z U^{-1}) d\eta \right. \\ &\quad \left. - \sum_{q=1,2} \eta (-1)^q D(e_2^\pm)^\top A (D_\eta U^0 + D_z U^{-1}) \Big|_{\eta=(-1)^q/2} \right) = 0. \end{aligned} \quad (2.20)$$

In the middle of formula (2.20), we used the simple identity

$$D(e_1^\pm) e_2^\pm = D(e_2^\pm \partial_\eta \eta) e_1^\pm, \quad (2.21)$$

the definition of the operator D_η , and the formula of integration by parts, whereas the relations (2.11) and (2.12) for $k = 0$ and $k = -1$ were used at the end of this formula. Taking into account the identities

$$D_\eta U^0 + D_z U^{-1} = (D_\eta \mathcal{X}^\pm + \mathcal{Y}^\pm) \mathcal{D}(\partial_z) w \quad (2.22)$$

we write the first ($i = 1$) solvability condition in (2.13) as the ordinary differential equation

$$-(e_1^\pm)^\top D_z^\top \int_{-1/2}^{1/2} A(D_\eta \mathcal{X}^\pm(\eta, z) + \mathcal{Y}^\pm(\eta)) d\eta \mathcal{D}(\partial_z) w^\pm(z) = 0. \quad (2.23)$$

Another equation for $w^\pm$ is obtained from the second ($i = 2$) solvability condition for the problems (2.11) and (2.12) with $k = 2$ and right-hand sides

$$\begin{aligned} F^2 &= D_\eta^\top A D_z U^1 + D_z^\top A(D_\eta U^1 + D_z U^0) + f_2^{0,\pm}, \\ G^{2,q} &= (-1)^q D(e_2^\pm)^\top A D_z U^1. \end{aligned} \quad (2.24)$$

The term $f_2^{0,\pm}$ is taken from the presentation (2.10).

As in the case (2.20), we have

$$\begin{aligned} 0 &= (e_2^\pm)^\top \left(\int_{-1/2}^{1/2} F^2 d\eta + \sum_{q=0,1} G^2 \right) = \frac{\partial}{\partial z} \int_{-1/2}^{1/2} ((D_\eta(\eta e_1^\pm))^\top A(D_\eta U^1 + D_z U^0) + f_2^{0,\pm}) d\eta \\ &= -(e_1^\pm)^\top \frac{\partial}{\partial z} \left(\int_{-1/2}^{1/2} (\eta D_\eta^\top A(D_\eta U^1 + D_z U^0) - f_2^{0,\pm}) d\eta \right. \\ &\quad \left. - \sum_{q=1,2} (-1)^q \eta D(e_2^\pm)^\top A(D_\eta U^1 + D_z U^0) \Big|_{\eta=(-1)^q/2} \right) \\ &= (e_1^\pm)^\top \frac{\partial}{\partial z} \left(\int_{-1/2}^{1/2} \eta D_z^\top A(D_\eta U^0 + D_z U^{-1}) d\eta \right) + \mathbf{F}_2^\pm. \end{aligned} \quad (2.25)$$

Moreover,

$$\mathbf{F}_2^\pm(z) = \int_{-1/2}^{1/2} f_2^{0,\pm}(z) d\eta = f_2^{0,\pm}(z), \quad (2.26)$$

and the equality (2.25) itself (without the middle part) takes the form

$$-(e_1^\pm)^\top \frac{\partial}{\partial z} \int_{-1/2}^{1/2} \eta_2 D_z^\top A(D_\eta \mathcal{X}^\pm(\eta) + \mathcal{Y}^\pm(\eta)) d\eta \mathcal{D}(\partial_z) w^\pm(z) = \mathbf{F}_2^\pm(z) \quad (2.27)$$

in view of (2.22). Since

$$\begin{aligned} (e_1^+)^\top D_z^\top &= (0, \partial_z, 0) \quad \text{on } \mathcal{Q}_+(h), \\ (e_1^-)^\top D_z^\top &= (\partial_z, 0, 0) \quad \text{on } \mathcal{Q}_-(h), \end{aligned}$$

we write the system (2.23), (2.27) in the matrix form

$$\mathbf{M}(\partial_z) w^\pm(z_\pm) := \mathcal{D}^\top(-\partial_z) M^\pm \mathcal{D}(\partial_z) w^\pm(z_\pm) = \mathbf{F}^\pm(z_\pm), \quad z_\pm \in \Upsilon_\pm := (0, l_\pm), \quad (2.28)$$

where formula (2.16) is taken into account. Here, $\mathbf{F}^\pm = (0, \mathbf{F}_2^\pm)^\top$ and $M^\pm$ are the number (2×2) -matrices; moreover, it is known (cf., for example, [7]) that

$$\begin{aligned} M^\pm &= \int_{-1/2}^{-1/2} \mathcal{Y}^\pm(\eta)^\top A(D_\eta \mathcal{X}^\pm(\eta) + \mathcal{Y}^\pm(\eta)) d\eta \\ &= \int_{-1/2}^{1/2} (D_\eta \mathcal{X}^\pm(\eta) + \mathcal{Y}^\pm(\eta))^\top A(D_\eta \mathcal{X}^\pm(\eta) + \mathcal{Y}^\pm(\eta)) d\eta. \end{aligned} \quad (2.29)$$

The last equality holds in view of (2.18), and the Gram matrix in (2.29) is symmetric and positive definite.

Remark 2.1 (cf. [7, Section 4.2]). If the entries of the matrix A are independent of the fast variable η , then the solutions $\mathcal{X}^\pm$ to the problems (2.18) and, consequently, the entries $M_{ij}^\pm$ of the matrices $M^\pm$ are explicitly expressed. Assume that A_{ij} are entries of the matrix A and m_{1j} are their minors, i.e.,

$$m_{11} = A_{22}A_{33} - A_{23}A_{32}, \quad m_{12} = A_{31}A_{23} - A_{21}A_{33}, \quad m_{13} = A_{21}A_{32} - A_{22}A_{31}.$$

Then

$$\mathcal{X}^\pm(\eta) = \begin{pmatrix} \sqrt{2}\eta m_{11}^{-1}m_{13} & -\eta^2 m_{11}^{-1}m_{13} \\ \eta m_{11}^{-1}m_{12} & -\frac{\sqrt{2}}{2}\eta^2 m_{11}^{-1}m_{12} \end{pmatrix}, \quad M^\pm = \frac{|A|}{m_{11}} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{6} \end{pmatrix}.$$

2.4. Boundary conditions for the resulting system

For the system (2.28) we impose boundary conditions at the points $z_\pm = l_\pm$ and transmission conditions at the point $z_\pm = 0$. As usual (cf., for example, [14, Section 3.3]), the clamping condition (2.1) on the node $\mathcal{G}_h^\pm$ generates the Dirichlet condition (1.9). At the point P_0 , for an node *fixed in two directions* x_1 and x_2 (in the terminology of [14]) the condition (1.10) holds. Furthermore, the “rotation” (i.e. the derivative with respect to transverse components of w in the longitudinal direction) is the same at the endpoints of the beams joined in some node, i.e., (1.12) holds at P_0 .

We define three more conditions, *intrinsic* in the terminology of [21] and closing the problem (2.28), (1.9)–(1.12). For this purpose we consider variational setting of the problem. We introduce the space $\mathbf{H}_0$ of functions

$$\mathbf{v} = (v_1^-, v_2^-, v_1^+, v_2^+) \in H^1(\Upsilon_-) \times H^2(\Upsilon_-) \times H^1(\Upsilon_+) \times H^2(\Upsilon_+) \quad (2.30)$$

satisfying the Dirichlet conditions (1.9) at the points $P_\pm$ and the stable transmission conditions (1.10) and (1.12) at the point P_0 . By a *weak solution* to the resulting problem we mean an element $(w_1^-, w_2^-, w_1^+, w_2^+)$ of the space $\mathbf{H}_0$ satisfying the integral identity

$$\mathbf{E}(w, v) := \sum_{\pm} (M\mathcal{D}(\partial_z)w^\pm, \mathcal{D}(\partial_z)v^\pm)_{\Upsilon_\pm} = \sum_{\pm} (\mathbf{F}_2^\pm, v_2^\pm)_{\Upsilon_\pm} \quad (2.31)$$

for any test “function” $\mathbf{v} \in \mathbf{H}_0$. As usual (cf. [24, 21]), the identity (2.31) is obtained by integrating with respect to $z_{\pm} \in \Upsilon_{\pm}$ Equations (2.28) multiplied by test vector-valued functions $v^{\pm}$ and using the Green formula

$$\begin{aligned} \sum_{\pm} (M \mathcal{D}(\partial_z) w^{\pm}, \mathcal{D}(\partial_z) v^{\pm})_{\Upsilon_{\pm}} &= \sum_{\pm} (\mathcal{D}^{\top}(-\partial_z) M^{\pm} \mathcal{D}(\partial_z) w^{\pm}, v^{\pm})_{\Upsilon_{\pm}} \\ &- \sum_{\pm} (M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(0), v_1^{\pm}(0) e_1^{\pm}) - (M^{+} \mathcal{D}(\partial_z) w^{+}(0), \partial_z v_2^{+}(0) e_2^{+}) \\ &- (M^{-} \mathcal{D}(\partial_z) w^{-}(0), \partial_z v_2^{-}(0) e_2^{-}). \end{aligned} \quad (2.32)$$

By (1.10) and (1.12), some terms outside the integral sign in this identity vanish, whereas the remaining terms vanish if the following three conditions hold at the point P_0 :

$$\begin{aligned} (e_1^{-})^{\top} M^{-} \mathcal{D}(\partial_z) w^{-}(0) &= 0, \\ (e_1^{+})^{\top} M^{+} \mathcal{D}(\partial_z) w^{+}(0) &= 0, \\ (e_2^{+})^{\top} M^{+} \mathcal{D}(\partial_z) w^{+}(0) + (e_2^{-})^{\top} M^{-} \mathcal{D}(\partial_z) w^{-}(0) &= 0. \end{aligned} \quad (2.33)$$

The first two conditions mean that the longitudinal forces vanish at coincident ends of beams, whereas the third condition means the continuity of bending moment. These relations, together with (1.10) and (1.12), form the necessary six transmission conditions at the point P_0 and close the boundary value problem for the system (2.28).

Proposition 2.1 (cf. [21]). *Assume that $k = 0, 1, 2, \dots$ and $\mathbf{F}_2^{\pm} \in H^{k-2}(\Upsilon_{\pm})$. Then there exists a unique solution to the problem (2.28), (1.9)–(1.12), (2.33) and the following estimate holds:*

$$\sum_{\pm} \|w^{\pm}; H^{k+2}(\Upsilon_{\pm}) \times H^{k+1}(\Upsilon_{\pm})\| \leq c_k \sum_{\pm} \|\mathbf{F}_2^{\pm}; H^{k-2}(\Upsilon_{\pm})\|. \quad (2.34)$$

We note that the unique solvability of the variational problem (2.31) follows from the Riesz representation theorem, whereas the smoothness improvement of the solution (the estimate (2.34) for $k \geq 2$) follows from the general results of [21, Chapter 2], in particular, Remark 2.9.5 concerning problems with transmission conditions.

3 Junction of Two Beams with Free Ends

3.1. Statement and solvability of the problem

We consider the system of equations (1.6) with the Neumann condition (2.2) on the boundary $\partial \mathcal{Q}(h)$. The weak statement of the problem

$$\mathcal{E}(u, v; \mathcal{Q}(h)) = (f, v)_{\mathcal{Q}(h)}, \quad v \in H^1(\mathcal{Q}(h))^2, \quad (3.1)$$

is considered in the Sobolev space $H^1(\mathcal{Q}(h))^2$ without any stable conditions on the boundary of the junction $\mathcal{Q}(h)$.

Since only a rigid displacement $\mathbf{d}b$ that is defined by a column $b \in \mathbb{R}^3$ and the matrix

$$\mathbf{d}(x) = \begin{pmatrix} 1 & 0 & \frac{\sqrt{2}}{2} x_2 \\ 0 & 1 & -\frac{\sqrt{2}}{2} x_1 \end{pmatrix} \quad (3.2)$$

satisfies the homogeneous ($f = 0$) problem (3.1), the Fredholm alternative provides three compatibility conditions in the problem (3.1):

$$\int_{Q^\pm(h)} \mathbf{d}^\top(x) f(h, x) dx = 0 \in \mathbb{R}^3. \quad (3.3)$$

As in the case of Lemma 2.1, the following known assertion holds.

Lemma 3.1. *If $f \in L_2(Q(h))^2$ and the conditions (3.3) are satisfied, then there exists a weak solution $u \in H(Q(h))^2$ to the problem (1.6), (2.2), defined up to a term $\mathbf{d}(x)b$, $b \in \mathbb{R}^3$. If the orthogonality conditions*

$$\int_{\mathcal{G}_h^0} \mathbf{d}^\top(x) u(h, x) dx = 0 \in \mathbb{R}^3 \quad (3.4)$$

hold, then the solution u is unique and satisfies the estimate (2.5).

3.2. New asymptotic ansats on beams

Recalling the comments in Subsection 1.4, we note that the asymptotic ansatz (2.6), (2.7) should be corrected since we need to take into account that the beams $Q_\pm(h)$ with free ends meet at a nonzero angle.

First of all, according to the general procedure [14, 16], we modify the formula for the leading term of the asymptotics

$$U^{-2}(z_\pm) = w_2^\pm(z_\pm) e_2^\pm + a_1^\pm e_1^\pm. \quad (3.5)$$

The introduction of a constant term (a longitudinal displacement of beams as a rigid body) to (3.5) does not affect the next following terms of the ansatz since the lower order terms U^k , $k = 0, 1$, in the previous ansatz (2.6) contain only the derivatives $\partial_z^{1+k} w_1^\pm$. Furthermore, modifying (3.5), we can impose the additional condition (3.7) on the functions $w_1^\pm$ in the formulas

$$\begin{aligned} U^{-1}(z_\pm, \eta_\pm) &= (w_1^\pm(z_\pm) - \eta_\pm \partial_z w_2^\pm(z_\pm)) e_1^\pm + W_2^\pm(z_\pm) e_2^\pm, \\ U^0(z_\pm, \eta_\pm) &= \mathcal{X}^\pm(\eta_\pm) \mathcal{D}(\partial_z) w^\pm(z_\pm) + (W_1^\pm(z_\pm) - \eta_\pm \partial_z W_2^\pm(z_\pm)) e_1^\pm, \end{aligned} \quad (3.6)$$

written with taking into account the lower order asymptotic terms (1.11).

Thus, in the new ansatz (2.6), (3.5), (3.6), there are three (not only $w_1^\pm$; cf. (2.7)) components of the longitudinal displacement of beams: $h^{-2} a_1^\pm e_1^\pm$, $h^{-1} w_1^\pm(z) e_1^\pm$, and $h^0 W_1^\pm(z) e_1^\pm$. To construct an asymptotic conglomerate (cf. comments to formula (1.11)), we need to consider each component separately.

With $\mathbf{w}^\pm = (\mathbf{w}_1^\pm, \mathbf{w}_2^\pm)^\top$ we associate two expressions $\mathbf{w}_1^\pm(0) = \mathbf{g}_1^\pm$ and $\partial_z \mathbf{w}_1^\pm(0) = \mathbf{g}_2^\pm$ at the point P_0 . It is clear that $\mathbf{g}_2^\pm = 0$, but $\mathbf{g}_1^\pm \neq 0$ in the first case $\mathbf{w}^\pm = (a_1^\pm, 0)^\top \neq 0 \in \mathbb{R}^2$. For $\mathbf{w}^\pm = w^\pm$, in addition to the first homogeneous (i.e., $\mathbf{g}_2^\pm = 0$) boundary condition (2.33), we impose the homogeneous ($\mathbf{g}_1^\pm = 0$) condition

$$\mathbf{w}_1^\pm(0) = 0, \quad (3.7)$$

which is the normalization, but not boundary condition. Finally, in the case $\mathbf{w}^\pm = W^\pm$, we assume that $\mathbf{g}_2^\pm \neq 0$; moreover, this lower order component of the asymptotic conglomerate (1.11) transforms the longitudinal force at the end of $Q_\pm(h)$ to the transverse force on the

coincident endpoint of $\mathcal{Q}_\mp(h)$ (cf. discussion in Subsections 1.4 and 3.4). The last modification is related to the matrix-valued functions $\mathcal{X}^\pm$ solving the problems (2.18). Namely, we take into account the arbitrariness (a rigid displacement) in the choice of a solution to the problem (2.18) and write this matrix in the form

$$\mathcal{X}^\pm(\eta) = \tilde{\mathcal{X}}^\pm(\eta) + e_i \mathbf{c}^{(\pm)}, \quad (3.8)$$

where $\mathbf{c}^{(\pm)} = (c^{(\pm)}, 0)$ and the particular solution $\tilde{\mathcal{X}}^\pm$ satisfies the orthogonality condition (2.19). The additional term does not affect the lower order terms of the expansion (2.6), but the constants $\mathbf{c}^{(\pm)}$ will be chosen to facilitate the formulation and derivation of the transmission conditions (cf. Subsection 4.6). They are also required to justify the asymptotics in Section 5.

3.3. Construction of additional terms of asymptotic ansatz

Repeating the computations of Subsection 2.3, we obtain the system (2.28) for $w^\pm$. Similarly, for $W^\pm = (W_1^\pm, W_2^\pm)$ we obtain the system of equations with the same matrix differential operator. The auxiliary vector $W^\pm$ is introduced to the asymptotic ansatz (3.6) in order to guarantee the solvability of the problem for $w^\pm$. Namely, for the right-hand side of the system

$$\mathbf{M}(\partial_z)W^\pm(z_\pm) = \mathbf{F}^\pm(z_\pm), \quad z_\pm \in \Upsilon_\pm,$$

we take the leading vector $\mathbf{F}^\pm$ of the longitudinal force on the beam $\mathcal{Q}_\pm(h)$:

$$\mathbf{F}^\pm = (\mathbf{C}_\pm, 0)^\top, \quad \mathbf{C}_\pm = l_\pm^{-1} \int_0^{l_\pm} f_1^{0,\pm}(z) dz \quad (3.9)$$

(cf. Subsections 3.6 and 3.7 and formulas (3.32), (3.39), (3.40)).

We can set $W_2^\pm = 0$. Then the functions $W_1^\pm$ satisfy the equation

$$-M_{11}\partial_z^2 W_1^\pm(z) = \mathbf{C}_\pm, \quad z \in \Upsilon_\pm. \quad (3.10)$$

We impose the boundary conditions

$$\begin{aligned} M_{11}\partial_z W_1^\pm(0) &= -l_\pm \mathbf{C}_\pm, \\ M_{11}\partial_z W_1^\pm(l_\pm) &= 0 \end{aligned} \quad (3.11)$$

such that the problem (3.10), (3.11) is solvable and has the solution

$$W_1^\pm(z_\pm) = \mathbf{C}_\pm(-z_\pm^2/2 + l_\pm z). \quad (3.12)$$

3.4. The boundary value problem for boundary layer in a neighborhood of the node $\mathcal{G}_h^0$

We introduce the stretched coordinates $\tau = (\tau_1, \tau_2) = h^{-1}(x_1, x_2)$. Making the change of variables $x \mapsto \tau$ and setting $h = 0$, we transform the junction $\mathcal{Q}(h)$ to the infinite domain $\Pi = G_1^0 \cup \varpi^- \cup \varpi^+$ (cf. Figure 5) formed by the node G_1^0 and two unit half-strips

$$\begin{aligned} \varpi^+ &= \{\tau : |\tau_1| < 1/2, \tau_2 > -1/2\}, \\ \varpi^- &= \{\tau : |\tau_2| < 1/2, \tau_1 > -1/2\}. \end{aligned} \quad (3.13)$$

We denote by $\gamma_i(h)$ the ends of $\mathcal{Q}_\pm(h)$ separating the beams from the node G_h^0 ,

$$\gamma_i(h) = \{(x_1, x_2) : x_i = 2h, |x_{3-i}| < h/2\}, \quad i = 1, 2, \quad (3.14)$$

and by γ_i their representations in the coordinates τ .

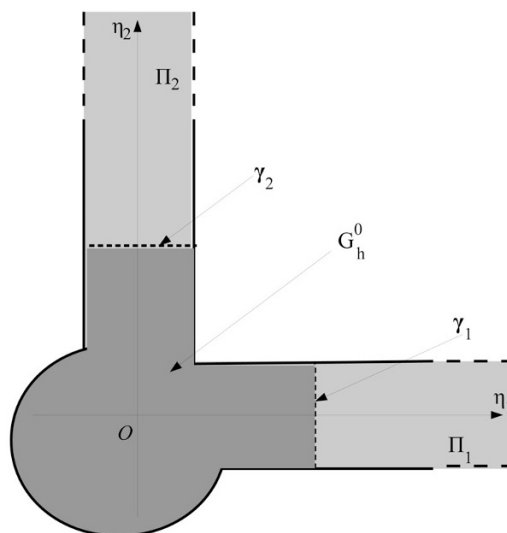


FIGURE 5. The domain Π for describing the boundary layer phenomenon.

To the asymptotic ansatz (2.6) we add the boundary layer type terms

$$\mathfrak{x}_0(x)(h^{-2}Z^{-2}(\tau) + h^{-1}Z^{-1}(\tau) + h^0Z^0(\tau) + \dots). \quad (3.15)$$

Here, $\mathfrak{x}_0$ is a smooth cut-off function that is equal to 1 on $\mathbb{B}^0(\mathbf{l}/3)$ and 0 outside $\mathbb{B}^0(2\mathbf{l}/3)$, $\mathbf{l} = \min\{l_-, l_+\}$. We look for Z^k as a solution to the problem of the theory of elasticity in the domain Π , obtained from (1.6), (2.2) by stretching coordinates $x \mapsto \tau$. Namely, we consider the system of equations

$$\begin{aligned} \mathbf{L}(\tau, \nabla_\tau)Z^k(\tau) &:= D(-\nabla_\tau)^\top A(\tau)D(\nabla_\tau)Z^k(\tau) = \mathbf{f}^k(\tau), \quad \tau \in \mathcal{G}_1^0, \\ \mathbf{L}(\nabla_\tau)Z^k(\tau) &= 0, \quad \tau \in \Pi \setminus \mathcal{G}_1^0, \end{aligned} \quad (3.16)$$

with the boundary conditions

$$\mathbf{B}(\tau, \nabla_\tau)Z^k(\tau) := D(\nu)^\top A(\tau)D(\nabla_\tau)Z^k(\tau) = 0, \quad \tau \in \partial\Pi, \quad (3.17)$$

and the jump conditions on the intervals γ_i (cf. Figure 5)

$$\begin{aligned} [Z^k(\tau)]_{\gamma_i}(\tau_{3-i}) &:= Z(\tau)|_{\tau_i=2+0} - Z(\tau)|_{\tau_i=2-0} = \mathbf{s}^{ki}(\tau_{3-i}), \\ [D(e_i)A(\tau)D(\nabla_\tau)Z^k(\tau)]_{\gamma_i}(\tau_{3-i}) &= \mathbf{g}^{ki}(\tau_{3-i}), \quad i = 1, 2. \end{aligned} \quad (3.18)$$

The jumps $\mathbf{s}^{ki}$, $\mathbf{g}^{ki}$ and the right-hand side of the system (3.16) on the node $\mathcal{G}_1^0$ are caused by the fact that the asymptotic ansatz (2.6), (3.5), (3.6) is located on the beams $\mathcal{Q}_\pm(h)$ and is not expanded to the node $\mathcal{G}_1^0$.

As usual (cf., for example, [24]), by a *weak solution* to the problem (3.16), (3.17), (3.18) we mean

$$Z^k = \mathbf{Z}^k + \mathbf{S}^k,$$

where $\mathbf{Z}^k \in H_{\text{loc}}^1(\overline{\Pi})^2$ and $\mathbf{S}^k \in H^1(\mathcal{G}_1^0)^2$ is a suitable extension of the displacement jumps $\mathbf{s}^{ki}$ to the node $\mathcal{G}_1^0$, i.e.,

$$\mathbf{S}^k = \mathbf{s}^{ki} \quad \text{on } \gamma_i, \quad \|\mathbf{S}^k; H^1(\mathcal{G}_1^0)\| \leq c \sum_{i=1,2} \|\mathbf{s}^{ki}; H^{1/2}(\gamma_i)\|, \quad (3.19)$$

where $H^{1/2}(\Xi)$ is the Sobolev–Slobodetskii space. The vector $\mathbf{Z}^k$ should satisfy the integral identity [24]

$$\mathcal{E}(\mathbf{Z}^k, V; \Pi) = \sum_{i=1,2} (\mathbf{g}^{ki}, V)_{\gamma_i} - \mathcal{E}(\mathbf{S}^k, V; \Pi) + (\mathbf{f}^{ki}, V)_{\Pi}$$

for all test (infinitely differentiable and compactly supported) vector-valued functions $V \in C_c^\infty(\overline{\Pi})^2$. The following assertion can be extracted, for example, from the surveys [25, 26] or the monograph [27].

Proposition 3.1. *Assume that $\mathbf{g}^{ki} \in L_2(\gamma_i)$ and $\mathbf{f}^k \in L_2(\mathcal{G}_1^0)$ satisfy the condition*

$$(\mathbf{d}^\top, \mathbf{f})_{\mathcal{G}_1^0} + (\mathbf{d}^\top, \mathbf{g}^{k1})_{\gamma_1} + (\mathbf{d}^\top, \mathbf{g}^{k2})_{\gamma_2} = 0 \in \mathbb{R}^3, \quad (3.20)$$

where $\mathbf{d}$ is defined by (3.2). Then the problem (3.16), (3.17), (3.18) has a unique solution

$$\mathbf{Z}^k(\tau) = \mathbf{S}^k(\tau) + \mathfrak{x}_+(\tau) \mathbf{d}(\tau) \mathbf{b} + \tilde{\mathbf{Z}}(\tau), \quad (3.21)$$

where $\mathbf{S}^k$ is defined by (3.19), $\mathfrak{x}_+$ is a smooth cut-off function, equal to 1 for $\tau > 2$ and 0 for $\tau \leq 1$, and the coefficient column $\mathbf{b} \in \mathbb{R}^3$ and exponentially decaying remainder $\exp(\beta_0|\tau|)\tilde{\mathbf{Z}} \in H^1(\Pi)$ with small $\beta_0 > 0$ satisfy the inequality

$$\|\exp(\beta_0|\tau|)\tilde{\mathbf{Z}}; H^1(\Pi)\| + |\mathbf{b}| \leq c \left(\|\mathbf{f}^k; L_2(\mathcal{G}_1^0)\| + \sum_{i=1,2} (\|\mathbf{g}^{ki}; L_2(\gamma_i)\| + \|\mathbf{s}^{ki}; H^1(\mathcal{G}_1^0)\|) \right),$$

where the constant c is independent of $\mathbf{f}^k$, $\mathbf{s}^{ki}$, and $\mathbf{g}^{ki}$.

Remark 3.1. The homogeneous ($\mathbf{f}^k = 0$ and $\mathbf{g}^{ki} = \mathbf{s}^{ki} = 0$) problem (3.16)–(3.18) has no nontrivial solutions decaying as $|\tau| \rightarrow \infty$, but it is satisfied by any rigid displacement. This fact is taken into account for constructing a unique solution (3.21): it vanishes at exponential rate at infinity in the half-strip ϖ^- (cf. (3.13)), but stabilizes to the rigid displacement $\mathbf{d}\mathbf{b}$ in the half-strip ϖ^+ . If $\mathbf{b} = 0 \in \mathbb{R}^3$, then the solution $\mathbf{Z}^k = \tilde{\mathbf{Z}}^k$ possesses the decay property which is attributable to the boundary layer.

3.5. The transmission conditions. Continuity of moment

Owing to the simple structure of the leading terms (3.5), (3.6), the right-hand sides $\mathbf{f}^k$ and $\mathbf{g}_i^k$ vanish for $k = -2, -1$ and the terms Z^{-2} , Z^{-1} of the expansion (3.15) are explicitly defined:

$$\begin{aligned} Z^{-2} &= a^- e_1 + a^+ e_2 \text{ on the node } \mathcal{G}_1^0, \\ Z^{-2} &= 0 \text{ on the half-strips } \varpi^\pm, \\ Z^{-1} &= e_1(w_1^-(0) - \tau_2 \partial_z w_2^-(0)) + e_2 \tau_1 \partial_z w_2^+(0) \text{ on the node } \mathcal{G}_1^0, \\ Z^{-1} &= e_1 w_1^-(0) - e_2 w_1^+(0) \text{ on the half-strip } \varpi^+, \\ Z^{-1} &= 0 \text{ on the half-strip } \varpi^-. \end{aligned}$$

We expand the higher order terms (2.7) and (2.17) into the Taylor series with respect to $z_{\pm}$ and pass to the stretched coordinates. Collecting coefficients at the same powers of the small parameter h , we find

$$\begin{aligned} \mathbf{f}^{0i} &= 0, \\ \mathbf{s}^{0i} &= (e_i(\tau_i \partial_z w_1(0) + W_1(0) - \tau_i \tau_{3-i} \partial_z^2 w_2(0)) + e_{3-i}(\tau_i^2 \partial_z^2 w_2(0)/2) \\ &\quad + \mathcal{X}(\tau_{3-i}) \mathcal{D}(\partial_z) w(0))|_{\tau_i=2}, \\ \mathbf{g}^{0i} &= D(e_i) A(\tau) (D(e_{3-i} \partial_{\tau_{3-i}}) \mathcal{X}(\tau_{3-i}) + \mathcal{Y}(\tau_{3-i})) \mathcal{D}(\partial_z) w(0), \quad i = 1, 2. \end{aligned} \quad (3.22)$$

We omit the superscripts “−” and “+” corresponding to $i = 1$ and $i = 2$ in the notation of the matrices $\mathcal{X}, \mathcal{Y}$ and vector w .

From the compatibility conditions (3.20) we obtain three transmission conditions on the components of $w^{\pm}$. We introduce these conditions by the simple identities:

$$e_{3-i}^{\top} D(e_i)^{\top} = e_i D(e_{3-i} \partial_{\tau_{3-i}})^{\top} \tau_{3-i}, \quad i = 1, 2, \quad (3.23)$$

and computations similar to (2.20) and (2.25). We consider the first two components of the vector (3.20). Multiplying $\mathbf{g}^{i0}$ by e_{3-i} , integrating with respect to γ_i , and taking into account (3.23), we find

$$\begin{aligned} e_{3-i}^{\top} \mathbf{g}^{i0} &= \int_{-1/2}^{1/2} e_i^{\top} D(e_{3-i} \partial_{\tau_{3-i}})^{\top} \tau_{3-i} A(\tau) \left(D(e_{3-i} \partial_{\tau_{3-i}}) \mathcal{X}^{\pm}(\tau_{3-i}) \right. \\ &\quad \left. + \mathcal{Y}^{\pm}(\tau_{3-i}) \right) \mathcal{D}(\partial_z) w^{\pm}(0) d\tau_{3-i} = -e_i^{\top} \int_{-1/2}^{1/2} \tau_{3-i} D(e_{3-i} \partial_{\tau_{3-i}})^{\top} \\ &\quad \times A(\tau) (D(e_{3-i} \partial_{\tau_{3-i}}) \mathcal{X}^{\pm}(\tau_{3-i}) + \mathcal{Y}^{\pm}(\tau_{3-i})) \mathcal{D}(\partial_z) w^{\pm}(0) d\tau_{3-i} \\ &\quad + e_i^{\top} \sum_{k=1,2} \eta(-1)^k \tau_{3-i} D(e_{3-i})^{\top} A(\tau) (D(e_{3-i} \partial_{\tau_{3-i}}) \mathcal{X}^{\pm}(\tau_{3-i}) \\ &\quad + \mathcal{Y}^{\pm}(\tau_{3-i})) \mathcal{D}(\partial_z) w^{\pm}(0) |_{\tau_{3-i}=(-1)^k/2} = 0. \end{aligned} \quad (3.24)$$

The last equality follows from the problem (2.18) for $\mathcal{X}^{\pm}$ since $\tau_{3-i} = \eta$ on γ_i . On the other hand, multiplying $\mathbf{g}^{i0}$ by e_i , integrating with respect to γ_i , and taking into account (2.29), we obtain the formula

$$\begin{aligned} e_i^{\top} \mathbf{g}^{i0} &= \int_{-1/2}^{1/2} e_i^{\top} D(e_i)^{\top} A(\tau) (D(e_{3-i} \partial_{\tau_{3-i}}) \mathcal{X}^{\pm}(\tau_{3-i}) \\ &\quad + \mathcal{Y}^{\pm}(\tau_{3-i})) \mathcal{D}(\partial_z) w^{\pm}(0) d\tau_{3-i} = (e_1^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(0). \end{aligned}$$

Consequently,

$$(e_1^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(0) = 0. \quad (3.25)$$

The last (the third column of the matrix (3.2)) solvability condition (3.20) can be transformed, as above, to the form

$$(e_2^+)^{\top} M^+ \mathcal{D}(\partial_z) w^+(0) + (e_2^-)^{\top} M^- \mathcal{D}(\partial_z) w^-(0) = 0. \quad (3.26)$$

It coincides with the last transmission condition in (2.33) on the node $\mathcal{G}_0(h)$ and implies the continuity of moment.

3.6. The transmission conditions. Equilibrium of the node $\mathcal{G}_0(h)$

It remains to obtain two transmission conditions at the point $z_{\pm} = 0$. For this purpose we consider the problem for Z^1 . As above, the right-hand sides

$$\mathbf{g}^{1i}(\tau) = D(e_i)A(\tau)D(\nabla_{\tau}) \sum_{n=0}^3 \frac{1}{n!} \tau_i^n \partial_z^n U^{1-n}(0, \tau_{3-i}) \Big|_{\tau_i=1}, \quad i = 1, 2,$$

are found by the Taylor formula for terms of the ansatz (2.6) (cf. (3.22)). Furthermore, $\mathbf{f}^{1i} = 0$. The vector-valued function U^1 is a solution to the problem (2.11), (2.12) with

$$\begin{aligned} F^1 &= (D_{\eta}^{\top} A D_z \mathcal{X}^{\pm} + D_z^{\top} A (D_{\eta} \mathcal{X}^{\pm} + D_z \mathcal{Y}^{\pm})) \mathcal{D}(\partial_z) w^{\pm} + D_{\eta}^{\top} A \mathcal{Y}^{\pm} \mathcal{D}(\partial_z) W^{\pm}, \\ G^{1,q} &= (-1)^q D(e_2^{\pm})^{\top} A (D_z \mathcal{X}^{\pm} \mathcal{D}(\partial_z) w^{\pm} + \mathcal{Y}^{\pm} \mathcal{D}(\partial_z) W^{\pm}), \quad q = 1, 2. \end{aligned}$$

Since the matrices $\mathcal{X}^{\pm}$ and $\mathcal{Y}^{\pm}$ are independent of z , we have

$$U^1(\eta, z) = \mathbf{X}^{\pm}(\eta) \mathcal{D}(\partial_z) \partial_z w^{\pm}(z) + \mathcal{X}^{\pm}(\eta) \mathcal{D}(\partial_z) W^{\pm}(z),$$

where $\mathbf{X}^{\pm}$ is a solution to the problem

$$\begin{aligned} L^0 \mathbf{X}^{\pm} &= D_{\eta}^{\top} A D_z \mathcal{X}^{\pm} + D_z^{\top} A (D_{\eta} \mathcal{X}^{\pm} + D_z \mathcal{Y}^{\pm}), \\ B^{0,q} \mathbf{X}^{\pm} &= (-1)^q D(e_2^{\pm})^{\top} A D_z \mathcal{X}^{\pm}, \quad q = 1, 2. \end{aligned}$$

The matrix $\mathbf{X}^{\pm}$ does not affect the coefficients of the operator $\mathbf{M}$ of the resulting system of equations and the transmission conditions. It is possible to avoid this influence by a suitable choice of the constant vector $\mathbf{c}^{(\pm)}$ in the representation (3.8) of the matrix $\mathcal{X}^{\pm}$ (cf. (3.29)). We consider the compatibility conditions

$$(e_i^{\top}, \mathbf{g}^{1i})_{\gamma_i} + (e_i^{\top}, \mathbf{g}^{13-i})_{\gamma_{3-i}} = 0, \quad i = 1, 2, \quad (3.27)$$

in the problem for Z^1 . Taking into account the representation (3.8) of the matrices $\mathcal{X}^{\pm}$ and the formula $D(e_i \partial_{\tau_i}) \tau_i \Phi(z) = D(e_i) \Phi(z) =: D_i \Phi(z)$, we have

$$\begin{aligned} e_i^{\top} \mathbf{g}^{1i} &= \{e_i^{\top} D_i^{\top} A(\tau) ((D_i e_i \mathbf{c}^{(\pm)} + D_i \tilde{\mathcal{X}}^{\pm} + D(e_{3-i} \partial_{\tau_{3-i}}) \mathbf{X}^{\pm}) \partial_z \mathcal{D}(\partial_z) w \\ &\quad + (D(e_{3-i} \partial_{\tau_{3-i}}) \mathcal{X}^{\pm}(\tau_{3-i}) + \mathcal{Y}^{\pm}(\tau_{3-i})) \mathcal{D}(\partial_z) W)\} \Big|_{\tau_i=1}. \end{aligned} \quad (3.28)$$

Here, the symbols “−” and “+” correspond to the cases $i = 1$ and $i = 2$ respectively. We substitute (3.28) into (3.27) and choose $\mathbf{c}^{(\pm)}$ in the following way:

$$\begin{aligned} \mathbf{c}^{(\pm)} &= -(\mathbf{a}^{(i)})^{-1} \int_{\gamma_i} e_i^{\top} D_i^{\top} A(\tau) (D_i \tilde{\mathcal{X}}^{\pm} + D(e_{3-i} \partial_{\tau_{3-i}}) \mathbf{X}^{\pm}) d\tau_{3-i}, \\ \mathbf{a}^{(i)} &= e_i^{\top} D_i^{\top} A D_i e_i > 0. \end{aligned} \quad (3.29)$$

Then we have

$$(e_i^\top, \mathbf{g}_i^1)_{\gamma_i} = (e_1^\pm)^\top M^\pm \mathcal{D}(\partial_z) W^\pm(0). \quad (3.30)$$

Using the identity (3.23) and repeating the transformations (3.24) for the terms containing components of the vector $w^\pm$, we find

$$(e_{3-i}^\top, \mathbf{g}_i^1)_{\gamma_i} = (e_2^\pm)^\top M^\pm \partial_z \mathcal{D}(\partial_z) w^\pm(0). \quad (3.31)$$

Summarizing (3.30) with (3.31) and taking into account (3.11), we obtain the boundary conditions on $w^\pm$

$$(e_2^\pm)^\top M^\pm \partial_z \mathcal{D}(\partial_z) w^\pm(0) = M_{11} \partial_z W_1^\mp(0) = -l_\mp \mathbf{C}_\mp =: \varphi_\mp, \quad (3.32)$$

in addition to the conditions (1.12), (3.25), (3.26) at the point $z_\pm = 0$.

We impose conditions on $(w_1^\pm, w_2^\pm)$ at the points $z_\pm = l_\pm$. We set

$$(e_1^\pm)^\top M^\pm \mathcal{D}(\partial_z) w^\pm(l_\pm) = 0, \quad (3.33)$$

$$(e_2^\pm)^\top M^\pm \mathcal{D}(\partial_z) w^\pm(l_\pm) = 0, \quad (3.34)$$

$$(e_2^\pm)^\top M^\pm \partial_z \mathcal{D}(\partial_z) w^\pm(l_\pm) = 0. \quad (3.35)$$

The boundary conditions (3.33)–(3.35) are taken according to the Green formula (2.31): owing to these boundary conditions, the term outside the integral sign in (2.31) vanishes. The rotation continuity condition (1.12) is extended by the definition of the constant

$$a_1^\mp = w_2^\pm(l_\pm), \quad (3.36)$$

which guarantees the continuity of translational displacements (3.5). We emphasize that the boundary conditions (3.33)–(3.35) can be obtained by analyzing the boundary layer in neighborhoods of the node $\mathcal{G}_h^\pm$. Finally, we require the normalization conditions for $w_1^\pm$:

$$\int_{\Upsilon_\pm} w_1^\pm dz = 0,$$

which completes the definition of the vector-valued functions $w^\pm$ and constants $a_1^\pm$ defining the asymptotic conglomerates (1.11) and (1.13).

3.7. The resulting boundary value problem

Thus, the vectors $w^\pm$ satisfy the limit problem

$$\begin{aligned} \mathbf{M}(\partial_z) w^\pm(z_\pm) &= \mathbf{F}^\pm(z_\pm), \\ \mathbf{B}^\pm(\partial_z) w^\pm(l_\pm) &= 0, \\ \mathbf{B}_0^\pm(\partial_z) w^\pm(0) &= \mathbf{G}^\pm, \end{aligned} \quad (3.37)$$

where $\mathbf{B}^\pm$ and $\mathbf{B}_0^\pm$ are the boundary operators (3.36), (1.12), (3.25), (3.26), (3.32)–(3.35), whereas only those $\mathbf{G}^\pm$ differ from zero that are contained in (3.32). The homogeneous ($\mathbf{F}^\pm = 0$,

$\mathbf{G}^\pm = 0$) system (3.37) has only three linearly-independent solutions $(w_1^-, w_1^+, w_2^-, w_2^+, a^-, a^+)$, namely,

$$\begin{aligned}\mathbf{w}^1 &= (0, 0, 1, 0, 0, 1), \\ \mathbf{w}^2 &= (0, 0, 0, 1, 1, 0), \\ \mathbf{w}^3 &= (0, 0, z_-, z_+, 0, 0).\end{aligned}\tag{3.38}$$

Formulas (3.3) and (2.10) guarantee the compatibility conditions

$$\begin{aligned}0 &= \int_{\Upsilon_\pm} \mathbf{F}_2^+(z) dz + \boldsymbol{\varphi}^- = \int_{\mathbf{Q}_\pm} (e_1^\top f^{0,+}(z) + e_1^\top f^{0,-}(z)) dz, \\ 0 &= \int_{\Upsilon_\pm} \mathbf{F}_2^-(z) dz + \boldsymbol{\varphi}^+ = \int_{\mathbf{Q}_\pm} (e_2^\top f^{0,-}(z) + e_2^\top f^{0,+}(z)) dz,\end{aligned}\tag{3.39}$$

and

$$0 = \int_{\Upsilon_\pm} z_+ \mathbf{F}_2^+(z) dz - \int_{\Upsilon_\pm} z_- \mathbf{F}_2^-(z) dz = \int_{\mathbf{Q}_\pm} (x_1 e_2^\top f_2^{0,+}(z) - x_2 e_1^\top f_2^{0,-}(z)) dz\tag{3.40}$$

obtained from the Fredholm alternative. Thus, under the conditions

$$\int_{\Upsilon_\pm} w_2^\pm(z) dz = 0, \quad \sum_{\pm} \int_{\Upsilon_\pm} z_\pm \partial_\pm w_2^\pm(z) dz = 0,\tag{3.41}$$

the problem (3.37) has a unique solution.

Proposition 3.2 (cf. [21]). *Assume that $k = 0, 1, \dots$, $\mathbf{F}_2^\pm \in H^{k-2}(\Upsilon_\pm)$. Let (3.39) and (3.40) hold. Then there exists a unique solution to the problem (3.37) satisfying the conditions (3.41); moreover, the following estimate holds:*

$$\sum_{\pm} \|w^\pm; H^{k+2}(\Upsilon_\pm) \times H^{k+1}(\Upsilon_\pm)\| \leq c_k \sum_{\pm} (\|\mathbf{F}_2^\pm; H^{k-2}(\Upsilon_\pm)\| + |\mathbf{G}_\pm|).\tag{3.42}$$

4 Junction of Beams with Periodicity Conditions at Endpoints

4.1. Statement of the problem

We consider the boundary value problem for Equation (1.6) with periodicity conditions at the endpoints of the junction $\mathcal{Q}_0(h)$ of beams of equal length $l_+ = l_- = l$ (cf. Subsection 1.5 and Figure 4). As in the case $\mathcal{Q}(h)$, we describe the junction $\mathcal{Q}_0(h)$ as the union of beams $\mathcal{Q}_\pm(h)$ and nodes (1.2), (1.3). On the intervals

$$\begin{aligned}\delta_h^- &= \{(x_1, x_2) : x_2 = t, x_1 = l + t, t \in (-h\sqrt{2}/2, h\sqrt{2}/2)\}, \\ \delta_h^+ &= \{(x_1, x_2) : x_1 = t, x_2 = l + t, t \in (-h\sqrt{2}/2, h\sqrt{2}/2)\},\end{aligned}$$

we impose the periodicity conditions

$$\begin{aligned}u(h, x(t_0))|_{\delta_h^-} + u(h, x(t_0))|_{\delta_h^+} &= 0, \\ \partial_\nu u(h, x(t_0))|_{\delta_h^-} + \partial_\nu u(h, x(t_0))|_{\delta_h^+} &= 0,\end{aligned}\tag{4.1}$$

where ∂_ν denotes the derivative with respect to the normal ν on $\delta_h^\pm$. On the remaining part of the boundary $\gamma_0(h) := \partial\mathcal{Q}_0(h) \setminus (\delta_h^- \cup \delta_h^+)$, we impose the Neumann condition

$$B(h, x, \nabla_x)u(h, x) = 0, \quad x \in \gamma_0(h). \quad (4.2)$$

As above, a weak statement of the problem (1.6), (4.1), (4.2) is the integral identity (3.1). The rotation $(x_2, -x_1)$ does not satisfy the condition (4.1), i.e., the solution u to the homogeneous problem (1.6), (4.1), (4.2) corresponds only to the translational displacements. By the Fredholm alternative, for the problem (1.6), (4.1), (4.2) there are two compatibility conditions

$$\int_{\mathcal{Q}_0(h)} f_i(h, x) dx = 0, \quad i = 1, 2. \quad (4.3)$$

Lemma 4.1. *If $f \in L_2(\mathcal{Q}(h))^2$ and (4.3) holds, then there exists a weak periodic solution $u \in H(\mathcal{Q}(h))^2$ to the problem (1.6), (4.1), (4.2) defined up to a summand $\mathbf{c}_1 e_1 + \mathbf{c}_2 e_2$, $\mathbf{c}_i \in \mathbb{R}$. If*

$$\int_{\mathcal{Q}_0(h)} u_i(h, x) dx = 0, \quad i = 1, 2,$$

then the solution u is unique and satisfies the estimate (2.5).

4.2. The transmission conditions

As in Subsection 4.1, we look for the leading terms of the asymptotics by formulas (3.5) and (3.6). Repeating the computation of Subsection 2.3, we obtain two systems of equations (2.28) for the vectors $w^\pm$ and $W^\pm$. The right-hand sides are defined by (2.26) for the first system and by (3.9) for the second one. We impose the transmission conditions on $w^\pm$. By the periodicity conditions (4.1) and the choice (3.6) of the higher order terms of the asymptotic ansatz, we find

$$w_2^\pm(0) = w_2^\pm(l) = a_1^\mp. \quad (4.4)$$

We assume that w_1 satisfies the antiperiodicity condition

$$w_1^\pm(0) = -w_1^\pm(l). \quad (4.5)$$

As above, we set $W_2^\pm = 0$, but impose the antiperiodicity condition

$$W_1^\pm(0) = -W_1^\pm(l). \quad (4.6)$$

The rotation continuity condition (in particular, on external node identified in view of periodicity) leads, in addition to (1.12), to the equality

$$\partial_z w_2^+(l) = \partial_z w_2^-(l). \quad (4.7)$$

We assume that $W_1^\pm$ satisfies the dual intrinsic boundary conditions of (4.6):

$$M_{11} \partial_z W_1^\pm(l) + M_{11} \partial_z W_1^\pm(0) = 0. \quad (4.8)$$

Then we obtain the uniquely solvable problem (3.10), (4.8), (4.6) with the constant $\mathbf{C}_\pm$ defined by (3.9). The solution

$$W_1^\pm(z) = \mathbf{C}_\pm(-z^2 + lz)/2$$

(cf. (3.12) for $l = l_{\pm}$) satisfies the identities

$$\begin{aligned} -M_{11}\partial_z W_1^{\pm}(0) &= -l\mathcal{C}_{\pm}/2, \\ M_{11}\partial_z W_1^{\pm}(l) &= -l\mathcal{C}_{\pm}/2. \end{aligned}$$

As in Subsections 3.6 and 3.7, the condition of existence of a decaying solution to the boundary layer problem implies the identities (3.26), (3.25) and the following relation similar to (3.32):

$$(e_2^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(0) = M_{11} \partial_z W_1^{\mp}(0) = l\mathcal{C}_{\mp}/2. \quad (4.9)$$

In small neighborhoods of the nodes $\mathcal{G}_h^{\pm}$ with centers $P_{\pm}$ (i.e., $z_{\pm} = l$), the behavior of the solution to the problem on junctions of beams is also described by the boundary layer. Identifying (which is possible by periodicity) the points P_+ and P_- , we introduce the new “stretched” coordinates $(\tau_1^+, \tau_2^+) = (h^{-1}(l - x_1), -h^{-1}x_2)$, $(\tau_1^-, \tau_2^-) = (-h^{-1}x_1, -h^{-1}(l - x_2))$ and, passing to the limit as $h \rightarrow 0$, we obtain the limit problem which has been already studied in Subsection 3.4. The compatibility conditions in this problem lead to the identities

$$(e_1^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(l) = 0, \quad (4.10)$$

$$\begin{aligned} (e_2^+)^{\top} M^+ \mathcal{D}(\partial_z) w^+(l) + (e_2^-)^{\top} M^- \mathcal{D}(\partial_z) w^-(l) &= 0, \\ (e_2^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(l) &= M_{11} \partial_z W_1^{\pm}(l) = l\mathcal{C}_{\mp}/2. \end{aligned} \quad (4.11)$$

From (3.25), (4.10), (4.9), and (4.11) we obtain the identities

$$\begin{aligned} (e_1^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(0) + (e_1^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(l) &= 0, \\ (e_2^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(0) - (e_2^{\pm})^{\top} M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(l) &= l\mathcal{C}_{\mp}, \end{aligned} \quad (4.12)$$

which completes the problem for $w^{\pm}$.

Thus, in the case of a “periodic” junction of beams, the vectors $w^{\pm}$ satisfy the result problem

$$\begin{aligned} \mathbf{M}(\partial_z) w^{\pm}(z_{\pm}) &= \mathbf{F}^{\pm}(z_{\pm}), \quad z_{\pm} \in \Upsilon_{\pm}, \\ \mathbf{B}_0^{\pm}(\partial_z) w^{\pm}(z_{\pm}) &= 0, \quad z_{\pm} = 0, l_{\pm}, \end{aligned} \quad (4.13)$$

where $\mathbf{B}_0^{\pm}$ denotes the boundary operators (4.4), (4.5), (1.12), (4.7), (3.25), (3.26), (4.10), (4.12) and $\mathbf{G}^{\pm}$ is the set of right-hand sides that are different from zero only in (4.12).

Let $\mathbf{H}$ be the space of collections of functions (2.30) satisfying the conditions (4.4), (4.5), and (4.7). By a *weak solution* to the resulting problem we mean a vector-valued function $(w_1^-, w_2^-, w_1^+, w_2^+) \in \mathbf{H}$ that for any test “function” $\mathbf{v} \in \mathbf{H} = (v_1^-, v_2^-, v_1^+, v_2^+)$ satisfies the integral identity (2.31) obtained by integrating Equations (2.28) multiplied by test vector-valued functions $v^{\pm} \in \mathbf{H}$ and using the Green formula (cf. (2.32))

$$\begin{aligned} \sum_{\pm} (M \mathcal{D}(\partial_z) w^{\pm}, \mathcal{D}(\partial_z) v^{\pm})_{\Upsilon_{\pm}} &= \sum_{\pm} (\mathcal{D}^{\top}(-\partial_z) M^{\pm} \mathcal{D}(\partial_z), v^{\pm})_{\Upsilon_{\pm}} \\ &+ \sum_{\pm} (M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(l) - M^{\pm} \mathcal{D}(\partial_z) w^{\pm}(0), v_1^{\pm}(0) e_1^{\pm}) \\ &+ (M^+ \mathcal{D}(\partial_z) w^+(l) - M^+ \mathcal{D}(\partial_z) w^+(0), \partial_z v_2^+(0) e_2^+) \\ &+ (M^- \mathcal{D}(\partial_z) w^-(l) - M^- \mathcal{D}(\partial_z) w^-(0), \partial_z v_2^-(0) e_2^-). \end{aligned}$$

The system (3.37) has two linearly independent solutions $\mathbf{w}^1$ and $\mathbf{w}^2$ defined in the previous section (cf. (3.38)). By the Fredholm alternative, for the system (3.37) there are two solvability conditions expressed by (3.39). By (4.3) and (2.10), these conditions are satisfied.

Proposition 4.1 (cf. [21]). *Assume that $k = 0, 1, \dots$ and $\mathbf{F}_2^\pm \in H^{k-2}(\Upsilon_\pm)$. Let (3.39) hold. Then there exists a unique solution to the problem (4.13) satisfying the first two conditions (3.41); moreover, the estimate (3.42) holds.*

5 Justification of Asymptotics

5.1. The Korn inequality for a junction of two beams with free ends

We justify the asymptotics for a junction of two beams with free ends. First of all, we formulate an asymptotically sharp weighted anisotropic Korn inequality which takes the simplest form on the node $\mathcal{G}_h^0$. By the orthogonality conditions (3.4), we have the inequality

$$|||u; \mathcal{G}_h^0|||_0^2 := (h^{-2}\|u; L_2(\mathcal{Q}(h))\|^2 + \|\nabla u; L_2(\mathcal{Q}(h))\|^2) \leq c\mathcal{E}(u, u; \mathcal{Q}(h)), \quad (5.1)$$

which can be easily verified by stretching the coordinates (cf. [25]). For the remaining elements of $\mathcal{Q}(h)$ we specify the results of [14], where the required inequality was obtained for an arbitrary system of beams.

Using the results of [14], we transfer “individual” Korn inequalities from one element of the elastic junction to another. Since the beams $\mathcal{Q}_\pm(h)$ are attached to the nodes where (5.1) holds, we have the following inequality on these beams:

$$\|\mathbf{u}; \mathcal{Q}_\pm(h)\|_\pm \leq c(|||u; \mathcal{G}_h^0|||_0 + \mathcal{E}(u, u; \mathcal{Q}(h))^{1/2}) \quad (5.2)$$

(cf. [14, Lemma 2.2]), where the anisotropic norm on the left-hand side is defined by

$$\begin{aligned} \|\mathbf{u}; \mathcal{Q}_\pm(h)\|_\pm := & \left(\int_{\mathcal{Q}_\pm(h)} \left(\left(\frac{\partial u_1}{\partial x_1} \right)^2 + \left(\frac{\partial u_2}{\partial x_2} \right)^2 \right. \right. \\ & \left. \left. + h^2 \left(\left(\frac{\partial u_1}{\partial x_2} \right)^2 + \left(\frac{\partial u_2}{\partial x_1} \right)^2 \right) + u_i^2 + h^2 u_{3-i}^2 \right) dx \right)^{1/2}. \end{aligned} \quad (5.3)$$

Here, the subscripts $i = 1$ and $i = 2$ correspond to the beams $\mathcal{Q}_-(h)$ and $\mathcal{Q}_+(h)$ respectively. The nodes $\mathcal{G}_h^\pm$ are external ends of the beams $\mathcal{Q}_\pm(h)$, where (5.2) is already obtained. In view of [14, Lemma 2.3],

$$|||u; \mathcal{G}_h^\pm|||_\pm \leq c(\|\mathbf{u}; \mathcal{Q}_\pm(h)\|_\pm + \mathcal{E}(u, u; \mathcal{Q}(h))) \quad (5.4)$$

on $\mathcal{G}_h^\pm$, and the weighted anisotropic norm is expressed as

$$\begin{aligned} |||u; \mathcal{G}_h^\pm|||_\pm := & \left(\int_{\mathcal{Q}_\pm(h)} \left(\left(\frac{\partial u_1}{\partial x_1} \right)^2 + \left(\frac{\partial u_2}{\partial x_2} \right)^2 + h^1 \left(\left(\frac{\partial u_1}{\partial x_2} \right)^2 \right. \right. \right. \\ & \left. \left. + \left(\frac{\partial u_2}{\partial x_1} \right)^2 \right) + h^{-1} u_i^2 + h^1 u_{3-i}^2 \right) dx \right)^{1/2}. \end{aligned} \quad (5.5)$$

Summarizing (5.1), (5.2), (5.4), we obtain the total Korn inequality

$$\sum_{\pm} (\|\mathbf{u}; \mathcal{Q}_\pm(h)\|_\pm + |||u; \mathcal{G}_h^\pm|||_\pm) + |||u; \mathcal{G}_h^0|||_0 \leq c\mathcal{E}(u, u; \mathcal{Q}(h))^{1/2}. \quad (5.6)$$

The asymptotic exactness of this inequality was established in [14].

5.2. Global asymptotic approximation

We begin with a suitable partition of unity on $\Omega(h)$. We introduce small squares

$$\begin{aligned}\square_+^1(h) &= \{(x_1, x_2) : |x_1| < h/2, l_+ - 2h < x_2 < l_+ - h\}, \\ \square_-^1(h) &= \{(x_1, x_2) : |x_2| < h/2, l_- - 2h < x_1 < l_- - h\}\end{aligned}$$

in the nodes $\mathcal{G}_h^+$ and $\mathcal{G}_h^-$ respectively and the squares

$$\begin{aligned}\square_+^0(h) &= \{(x_1, x_2) : |x_1| < h/2, h < x_2 < 2h\}, \\ \square_-^0(h) &= \{(x_1, x_2) : |x_2| < h/2, h < x_1 < 2h\}\end{aligned}$$

in the square $\mathcal{G}_h^0$. Let χ_h^s , $s = 0, \pm$, be a smooth cut-off function with $\text{supp}\chi_h^s \subset \mathcal{G}_h^s$. Similarly, we introduce cut-off functions X_h^S , $S = +, -$, that are equal to 1 on $\mathcal{Q}_S(h)$ and 0 on $\mathcal{G}_h^S \setminus \square_S^1(h)$ and thereby on $\mathcal{G}_h^0 \setminus (\square_+^0(h) \cup \square_-^0(h))$. Furthermore, X_h^S depends only on the longitudinal variable z_S and

$$|\partial_z^p X_h^S(z)| \leq c_p h^{-p}, \quad |\nabla_z^p \chi_h^s(x)| \leq c_p h^{-p}, \quad p = 0, 1, \dots$$

The *global asymptotic approximation* $\mathbf{U}(h, x)$ is the sum of the asymptotic constructions $X_h^\pm(z)\mathcal{U}_X^\pm(h, x)$ on the beams $\mathcal{Q}_\pm(h)$ and $\chi_h^s(x)\mathcal{U}_\mathcal{G}^s(h, x)$ on the nodes $\mathcal{G}_h^s$. Here,

$$\mathcal{U}_X^\pm(h, x) = h^{-2}U^{-2}(z) + h^{-1}U^{-1}(\eta, z) + h^0U^0(\eta, z) + h^1U^1(\eta, z) \quad (5.7)$$

and

$$\begin{aligned}\mathcal{U}_\mathcal{G}^\pm(h, x) &= \chi_h^\pm(z)h^{-2}w_2^\pm(l_\pm) + h^{-1}(w_1^\pm(l_\pm) - \tau\partial_z w_2^\pm(l_\pm) + W_2^\pm(l_\pm)), \\ \mathcal{U}_\mathcal{G}^0(h, x) &= h^{-2}Z^{-2}(\tau) + h^{-1}Z^{-1}(\tau).\end{aligned} \quad (5.8)$$

Moreover, U^{-2} , U^{-1} are expressed by (3.5), (3.6) and

$$U^0(z, \eta) = \mathcal{X}^\pm(\eta, z)\mathcal{D}(\partial_z)w^\pm(z) + \mathcal{X}^\pm\mathcal{D}(\partial_z)W^\pm, \quad (5.9)$$

whereas Z^{-2} and Z^{-1} are introduced in Subsection 3.5. In (5.7) and below, we often omit the superscripts “ $\pm$ ” in order to shorten the notation. By Proposition 2.1, from the smoothness conditions (5.14) we obtain the relation $w^\pm \in H^4(\Upsilon_\pm) \times H^3(\Upsilon_\pm)$ and thereby we can find all terms of the asymptotic ansätze (5.7) and (5.8).

5.3. Computation of residuals

We extract and study residuals generated by $\mathcal{U}_X^\pm$ in the system (1.6). By (2.11),

$$\begin{aligned}L\mathcal{U}_X^\pm &= h^{-4}L^0U^{-2} + h^{-3}(L^0U^{-1} + L^1U^{-2}) + h^{-2}(L^0U^0 + L^1U^{-1} + L^2U^{-2}) \\ &\quad + h^{-1}(L^0U^1 + L^1U^0 + L^2U^{-1}) + h^0(L^2U^0 + L^1U^1) + h^1L^2U^1 =: \mathcal{F}_\pm.\end{aligned} \quad (5.10)$$

By the definition of U^k , the terms in (5.10) with factors h^k , $k < 0$, vanish, i.e.,

$$\mathcal{F}_\pm = h^0(L^1U^1 + L^2U^0) + h^1L^2U^1.$$

According to (2.12), the boundary conditions have the form

$$B\mathcal{U}_X^\pm = h^1B^{1,q}U^1 =: \mathcal{G}_\pm^q, \quad q = 1, 2. \quad (5.11)$$

The difference $\mathcal{R} = u - \mathbf{U}$ between the exact and approximate solutions satisfies the system

$$L\mathcal{R}(h, x) = f(h, x) - \mathcal{F}_\pm(h, x), \quad x \in \mathcal{Q}_\pm(h). \quad (5.12)$$

On the nodes, the difference $\mathcal{R}$ leaves a residual concentrated only in the sets $\square_\pm^k(h)$, $k = 0, 1$, where the cut-off functions are not constants. Namely,

$$L\mathcal{R}(h, x) = L(X_h^\pm(z_\pm) - 1)\mathcal{U}_X^\pm(h, x) + \mathcal{F}_\pm(h, x) + L(\chi_h^s(x))\mathcal{U}_G^s(h, x) \quad (5.13)$$

for $x \in \mathcal{G}_h^s$, $s = 0, \pm$. To estimate the remainder, we specify the representation (2.10) of the right-hand side f of the system (2.2) as follows:

$$f(h, x) = f^{0,\pm}(z_\pm) + \tilde{f}^\pm(h, x);$$

moreover,

$$f^{0,\pm} \in L_2(\Upsilon_\pm) \quad (5.14)$$

and

$$\|\tilde{f}^\pm; L_2(\Upsilon_\pm)\| \leq Ch. \quad (5.15)$$

We denote

$$\mathcal{N} := \sum_{\pm} (\|\tilde{f}^\pm; L_2(\Upsilon_\pm)\| + \|f^{0,\pm}; L_2(\Upsilon_\pm)\|). \quad (5.16)$$

We take the inner products of the systems (5.12) and (5.13) with $\mathcal{R}$ and integrate by parts over $\mathcal{Q}(h)$. By the identity $D(\nabla_x)\mathcal{U}_X^s(x) = 0$, $x \in \overline{\mathcal{G}_h^s}$, and formulas (2.4), (5.12), (5.13), (5.11),

$$\begin{aligned} \mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h)) &= \sum_{\pm} \left(-(\mathcal{G}_\pm, \mathcal{R})_{\Sigma_\pm} - (\mathcal{F}_\pm, \mathcal{R})_{\mathcal{Q}_\pm(h) \cup \square_\pm^0(h) \cup \square_\pm^1(h)} \right. \\ &\quad + (D(\nabla_x)^\top AD(\nabla_x)((X_\pm - 1)\mathcal{U}_X^\pm + \chi_\pm \mathcal{U}_X^\pm), \mathcal{R})_{\square_\pm^1(h)} \\ &\quad + (D(\nabla_x)^\top AD(\nabla_x)((X_\pm - 1)\mathcal{U}_X^\pm + \chi_0 \mathcal{U}_X^0, \mathcal{R}))_{\square_\pm^0(h)} + (f, \mathcal{R})_{\mathcal{Q}(h)} \\ &\quad - (D(\nu^{1\pm})^\top AD(\nabla_x)((X_\pm - 1)\mathcal{U}_X^\pm + \chi_\pm \mathcal{U}_X^\pm), \mathcal{R})_{\Sigma_\pm \cap \partial \square_\pm^1(h)} \\ &\quad \left. - (D(\nu^{0\pm})^\top AD(\nabla_x)((X_\pm - 1)\mathcal{U}_X^\pm + \chi_0 \mathcal{U}_X^0), \mathcal{R})_{\Sigma_\pm \cap \partial \square_\pm^0(h)} \right). \end{aligned} \quad (5.17)$$

Here, $\nu^{k\pm}$ is the outward normal on $\Sigma_\pm^k$ and $\Sigma_\pm = \Sigma_\pm^1 \cup \Sigma_\pm^2$. Integrating by parts, we remove the integral over the sides of the beams $\mathcal{Q}_\pm(h)$ from (5.17). As a result, we obtain the identities

$$\mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h)) = \sum_{\pm} \left(J_X^\pm + \tilde{I}^\pm + I_0^\pm + I_{01}^\pm + h^1 I_1^\pm + \sum_{k=0,1} (-1)^{k+1} J_{k\gamma}^\pm \right), \quad (5.18)$$

where

$$\begin{aligned} J_X^\pm &= (AD(\nabla_x)((1 - X_\pm)\mathcal{U}_X^\pm - \chi_\pm \mathcal{U}_X^\pm), D(\nabla_x)\mathcal{R})_{\square_\pm^1(h)} \\ &\quad + (AD(\nabla_x)((1 - X_\pm)\mathcal{U}_X^\pm - \chi_0 \mathcal{U}_X^0), D(\nabla_x)\mathcal{R})_{\square_\pm^0(h)}, \\ I_{01}^\pm &= h(D(\nu^{k\pm})^\top AD_z U^1, \mathcal{R})_{\Sigma_\pm} + (D_\eta AD_z U^1, \mathcal{R})_{\mathcal{Q}_\pm(h)}, \\ I_0^\pm &= (f^{0\pm}, \mathcal{R})_{\mathcal{Q}_\pm(h)} - (D_z^\top A(D_\eta U^1 + D_z U^0), \mathcal{R})_{\mathcal{Q}_\pm(h)}, \\ I_1^\pm &= (D_z^\top AD_z U^1, \mathcal{R})_{\mathcal{Q}_\pm(h)}, \quad \tilde{I}^\pm = (\tilde{f}^\pm, \mathcal{R})_{\mathcal{Q}_\pm(h)}, \\ J_{k\gamma}^\pm &= (D(e_1^\pm)^\top AD(\nabla_x)\mathcal{U}_X^\pm, \mathcal{R})_{\gamma_\pm^k(h)}. \end{aligned} \quad (5.19)$$

Finally, $\gamma_-^0(h) = \gamma_1(h)$, $\gamma_+^0(h) = \gamma_2(h)$ (cf. (3.14)) and

$$\begin{aligned}\gamma_-^1(h) &= \{(x_1, x_2) : x_1 = l_- - 2h, |x_2| < h/2\}, \\ \gamma_+^1(h) &= \{(x_1, x_2) : x_2 = l_+ - 2h, |x_1| < h/2\}.\end{aligned}$$

Our next goal is to estimate the terms (5.19) in the leading part of (5.18) by the quantity $\mathbf{c}\mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h))^{1/2}$, where the constant $\mathbf{c}$ depends only on the right-hand side f of (1.6).

5.4. Estimation of residuals on the beams $\mathcal{Q}_\pm(h)$

By the definition of the norm (5.3) and the Korn inequality (5.2),

$$|\tilde{I}^\pm| \leq ch \|\tilde{f}^\pm; L_2(\mathcal{Q}_\pm(h))\| \|\mathcal{R}; L_2(\mathcal{Q}_\pm(h))\| \leq c\mathcal{N} \|\mathcal{R}; \mathcal{Q}_\pm(h)\|_\pm \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h))^{1/2}.$$

By the condition (5.14) and Proposition 2.1, $w^\pm \in H^4(\Upsilon_\pm) \times H^3(\Upsilon_\pm)$, satisfies (2.34) with $k = 2$. Consequently, U^j satisfies the inclusion $U^j \in H^{2-j}(\Upsilon_\pm)^2$ and

$$\|U^j; H^{2-j}(\Upsilon_\pm)^2\| \leq c\mathcal{N}. \quad (5.20)$$

By (5.20),

$$\begin{aligned}h|I_1^\pm| &\leq ch \|D_z^\top AD_z U^1; L_2(\mathcal{Q}_\pm(h))\| \|\mathcal{R}; L_2(\mathcal{Q}_\pm(h))\| \\ &\leq c\mathcal{N} \|\mathcal{R}; L_2(\mathcal{Q}_\pm(h))\| \leq c\mathcal{N} \|\mathcal{R}; \mathcal{Q}_\pm(h)\|_\pm \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h))^{1/2}.\end{aligned}$$

The estimate $\|D_z^\top AD_z U^1; L_2(\mathcal{Q}_\pm(h))\| \leq c\mathcal{N}$ follows from (5.9). Integrating by parts, we find $I_{01}^\pm = -h(AD_z U^1, D_y \mathcal{R})_{\mathcal{Q}_\pm(h)}$, where $y = h\eta$ and $D_y = D(e_2^\pm \partial_y)$. Using (5.20) and the Korn inequality (5.2), we derive the estimate

$$|I_{01}^\pm| \leq c\mathcal{N} \|\mathcal{R}; \mathcal{Q}_\pm(h)\|_\pm \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h))^{1/2}.$$

To estimate $I_0^\pm$, we need the following assertion (cf. [14, 28]).

Lemma 5.1. *For $R \in H^1(\mathcal{Q}_\pm(h))^2$*

$$\|R_1; L_2(\mathcal{Q}_\pm(h))\|^2 + \|R_2 - \langle R_2 \rangle e_2^\pm; L_2(\mathcal{Q}_\pm(h))\|^2 \leq c\mathcal{E}(R; R; \mathcal{Q}_\pm(h)), \quad (5.21)$$

where

$$\langle R_2 \rangle(z) = h^{-1} \int_{-h/2}^{h/2} R_2(y, z) dy.$$

We recall that the second row of the system (2.28) is the right-hand side of the identities (2.25). In the case of a junction of two beams with free ends, we add to F^2 and $G^{2,q}$ in (2.24) the following terms generated by the lower order asymptotic conglomerate (1.11):

$$D_\eta^\top AD_z \mathcal{X}^\pm \mathcal{D}(\partial_z) W^\pm + D_z^\top AD_\eta \mathcal{X}^\pm \mathcal{D}(\partial_z) W^\pm + D_z^\top AD_z e_1^\pm (W_1^\pm + \eta \partial_z W_2^\pm) \quad (5.22)$$

and

$$D(\nu^{k\pm})^\top AD_z \mathcal{X}^\pm \mathcal{D}(\partial_z) W^\pm. \quad (5.23)$$

At the same time, the expressions (5.22) and (5.23) do not contribute to Equation (2.28), and it is easy to verify that the chain of identities (2.25) is preserved. Taking into account (5.22), (5.23) and multiplying the first equality in (2.25) by $\langle R_2 \rangle$, we get

$$0 = (D_z A(D_z U^0 + D_\eta U^1), \langle R_2 \rangle)_{\mathcal{Q}_\pm(h)} - (f^{0\pm}, \langle R_2 \rangle)_{\mathcal{R}_\pm}. \quad (5.24)$$

Subtracting from $I_0^\pm$ the relation (5.24) multiplied by h , we find

$$I_0^\pm = (D_z A(D_z U^0 + D_\eta U^1) - f^{0\pm}, \mathcal{R} - \langle R_2 \rangle e_2^\pm)_{\mathcal{Q}_\pm(h)}. \quad (5.25)$$

Applying the estimate (5.21) and the inequalities (5.20), (5.14) to (5.25), we get

$$|I_0^\pm| \leq c\mathcal{N}[\mathcal{R}; \mathcal{Q}_\pm(h)]_\pm \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h))^{1/2}.$$

5.5. Estimation of residuals on sides of small squares

We consider the last sum in (5.18). By (5.7), the definitions of U^k , $V_0^\pm$ in the asymptotic ansatz and the representation (2.8) of the operator $D(\nabla_x)$, we have

$$J_{k\gamma}^\pm = h^{-1}\mathcal{I}_{k\gamma}^\pm + h^0\mathcal{J}_{k\gamma}^\pm + h^1\mathcal{K}_{k\gamma}^\pm,$$

where

$$\mathcal{I}_{k\gamma}^\pm = (D(e_1^\pm)^\top A(D_\eta U^0 + D_z U^{-1}), \mathcal{R})_{\gamma_\pm^k(h)} = (D(e_1^\pm)^\top A(D_\eta \mathcal{X}^\pm + \mathcal{Y}^\pm) \mathcal{D}(\partial_z) w^\pm, \mathcal{R})_{\gamma_\pm^k(h)},$$

$$\mathcal{J}_{k\gamma}^\pm = (D(e_1^\pm)^\top A(D_z U^0 + D_\eta U^1), \mathcal{R})_{\gamma_\pm^k(h)},$$

$$\mathcal{K}_{k\gamma}^\pm = (D(e_1^\pm)^\top A D_z U^1, \mathcal{R})_{\gamma_\pm^k(h)}.$$

By the standard trace inequality

$$h^1 \|\mathcal{V}; L_2(\gamma_\pm^k(h))\|^2 \leq c(\|\mathcal{V}; L_2(\square_\pm^k(h))\|^2 + h^2 \|\nabla_x \mathcal{V}; L_2(\square_\pm^k(h))\|^2) \quad (5.26)$$

we have the estimate

$$h|\mathcal{K}_{k\gamma}^\pm| \leq ch \|AD_z U^1; L_2(\gamma_\pm^k(h))\| \|D(\nabla_x) \mathcal{R}; L_2(\gamma_\pm^k(h))\| \leq ch^{1/2} \mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h))^{1/2}.$$

On the square $\square_\pm^k(h)$, we represent the asymptotic remainder $\mathcal{R}$ as the sum

$$\mathcal{R}(h, x) = \mathcal{R}_\pm^{k\pm}(h, x) + \mathbf{d}_\pm^k(x) \overline{\mathcal{R}}^{k\pm}(h, x),$$

where

$$\begin{aligned} \mathbf{d}_\pm^0(x) &= \mathbf{d}(x), \quad \mathbf{d}_\pm^1(x) = \begin{pmatrix} 1 & 0 & \frac{\sqrt{2}}{2}x_2 \\ 0 & 1 & \frac{\sqrt{2}}{2}(l_\pm - x_1) \end{pmatrix}, \\ \overline{\mathcal{R}}^{k\pm}(h, x) &= \left(\int_{\square_\pm^k(h)} \mathbf{d}(x) \mathbf{d}(x)^\top dx \right)^{-1} \int_{\square_\pm^k(h)} \mathbf{d}(x)^\top \mathcal{R}(h, x) dx, \\ \int_{\square_\pm^k(h)} \mathcal{R}_\pm^{k\pm}(h, x) dx &= 0. \end{aligned} \quad (5.27)$$

Formula (3.2) for the matrix $\mathbf{d}$ shows that

$$\begin{aligned} h^{-1}\mathcal{I}_{k\gamma}^{\pm} + \mathcal{J}_{k\gamma}^{\pm} &= h^{-1} \sum_{q=0}^3 I_k^{q\pm} + \sum_{q=0}^3 J_k^{q\pm}, \\ I_k^{q\pm} &:= (D(e_1^{\pm})^{\top} A(D_{\eta}\mathcal{X}^{\pm} + \mathcal{Y}^{\pm})\mathcal{D}(\partial_z)w^{\pm}, R_q^{k\pm})_{\gamma_{\pm}^k(h)}, \\ J_k^{q\pm} &:= (D(e_1^{\pm})^{\top} A(D_z U^0 + D_{\eta} U^1), R_q^{k\pm})_{\gamma_{\pm}^k(h)}; \end{aligned} \quad (5.28)$$

moreover, $R_0^{k\pm} = \mathcal{R}_{\pm}^{k\pm}$, $R_1^{k\pm} = e_1^{\pm} \overline{\mathcal{R}}^{k\pm}$, $R_2^{k\pm} = e_2^{\pm} \overline{\mathcal{R}}^{k\pm}$, $R_3^{k\pm} = \sqrt{2}/2(x_2, -x_1)^{\top} \overline{\mathcal{R}}^{k\pm}$. We have $I_1^{q\pm} = 0$ in view of the identity $e_1^{\top} D^{\top}(e_1) = e_1^{\top} \mathcal{Y}(\eta)^{\top}$ which can be easily verified, the definition (2.29) of the matrix $\mathbf{M}$, and the boundary conditions (3.33), (3.25). Based on computations similar to (2.20) and the identity (2.21), we find $I_2^{q\pm} = 0$. From the identities

$$\begin{aligned} h^{-1} \frac{\sqrt{2}}{2} x_2 e_1^{\top} D^{\top}(e_1) &= (e_2^{-})^{\top} \mathcal{Y}^{-\top}(\eta^{-}), \\ h^{-1} \frac{\sqrt{2}}{2} x_1 e_2^{\top} D^{\top}(e_1) &= (e_2^{+})^{\top} \mathcal{Y}^{+\top}(\eta^{+}), \end{aligned}$$

the definition (2.29) of the matrix $\mathbf{M}$, the boundary (3.34) and transmission (3.26) conditions we obtain the identities $I_3^{1\pm} = 0$ and $I_3^{0\pm} + I_3^{0\mp} = 0$. Thus, the first sum in (5.28) contains only the terms $I_0^{q\pm}$. To estimate these terms, we use the inequalities (5.1) following from (5.27) and the trace inequality (5.26). We have

$$\begin{aligned} h^{-1}|I_0^{k\pm}| &\leq ch^{-1}h^{1/2}\mathcal{N}\|\mathcal{R}_{\pm}^{k\pm}; L_2(\gamma_{\pm}^k(h))\| \\ &\leq ch^{-1}h^{1/2}h^{-1/2}\mathcal{N}(\|\mathcal{R}_{\pm}^{k\pm}; L_2(\square_{\pm}^k(h))\|^2 + h^2\|\nabla_x \mathcal{R}_{\pm}^{k\pm}; L_2(\square_{\pm}^k(h))\|^2)^{1/2} \\ &= c\mathcal{N}|||\mathcal{R}_{\pm}^{k\pm}; \square_{\pm}^k(h)|||_0 \leq c\mathcal{N}\mathcal{E}(\mathcal{R}_{\pm}^{k\pm}, \mathcal{R}_{\pm}^{k\pm}; \square_{\pm}^k(h)) = c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \square_{\pm}^k(h))^{1/2}. \end{aligned}$$

The factor $h^{1/2}$ is caused by integration over the transverse cross-section of the square $\square_{\pm}^q$. Similarly, we derive the estimate

$$h^0|J_0^{k\pm}| \leq ch^1\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \square_{\pm}^k(h)).$$

Since the components of the third columns of the matrices $\mathbf{d}_{\pm}^k$ on $\gamma_{\pm}^k$ are of order $O(h^{1/2})$, we have

$$h^0|J_3^{k\pm}| \leq ch^1\mathcal{N}\|\mathcal{R}_{\pm}^{k\pm}; L_2(\square_{\pm}^k(h))\| \leq c\mathcal{N}|||\mathcal{R}; \mathcal{Q}_{\pm}(h)|||_{\pm} \leq c\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \square_{\pm}^k(h)).$$

By the usual trace inequality

$$\|e_1 \mathcal{V}; L_2(\gamma_{\pm}^k(h))\|^2 \leq c(\|e_1 \mathcal{V}; L_2(\square_{\pm}^k(h))\|^2 + \|\nabla_x \mathcal{V}; L_2(\square_{\pm}^k(h))\|^2)$$

and the inequality

$$\|e_1 \mathcal{V}; L_2(\square_{\pm}^k(h))\| \leq |||\mathcal{V}; (\square_{\pm}^k(h))|||_{\pm},$$

we obtain the estimate

$$h^0|J_1^{k\pm}| \leq ch^0\mathcal{N}\mathcal{E}(\mathcal{R}, \mathcal{R}; \square_{\pm}^k(h)).$$

Finally, by computations similar to (2.25), the boundary (3.35) and transmission (3.32) conditions, and the choice (3.29) of the constant $\mathbf{c}^{(\pm)}$, we obtain the identities $J_2^{1\pm} = 0$ and $J_2^{0\pm} + J_2^{0\mp} = 0$.

5.6. Estimation of residuals on small squares

Let us estimate the integrals $J_X^\pm$. By (5.8) and the identities $\chi_h + X_h = 1$, $\nabla\chi_h + \nabla X_h = 0$, we conclude that

$$J_X^\pm = \sum_{\pm} \sum_{q=0,1} (AD(\nabla)(1 - X_h^\pm))\tilde{\mathcal{U}}^{q\pm}, D(\nabla)\mathcal{R})_{\square_\pm^q}, \quad (5.29)$$

where

$$\begin{aligned} \tilde{\mathcal{U}}^1(z) &= h^{-2}e_2(w_2(z) - w_2(l) - (l - z)\partial_z w_2(l)) + h^{-1}(e_1(w_1(z) - w_1(l) \\ &\quad - \eta(\partial_z w_2(z) - \partial_z w_2(l))) + e_2(W_2(z) - W_2(l))) + h^0 U^0(l, \eta) + h^1 V^0(l, \eta), \\ \tilde{\mathcal{U}}^0(z) &= h^{-2}e_2(w_2(z) - w_2(0) - z\partial_z w_2(0)) + h^{-1}(e_1(w_1(z) - w_1(0) \\ &\quad - \eta(\partial_z w_2(z) - \partial_z w_2(0))) + e_2(W_2(z) - W_2(0))) + h^0 U^0(0, \eta) + h^1 V^0(0, \eta). \end{aligned}$$

We omit the subscripts $\pm$ in (5.30) in order to simplify the notation.

To estimate $J_X^\pm$, we use the following assertion which can be proved, for example, with the help of the Hardy inequality (cf. [7, Lemma 1.2.4]).

Lemma 5.2. *If $\mathcal{V} \in \dot{H}^1(\mathcal{Q}_\pm(h))$ and $\mathcal{W} \in H^1(\mathcal{Q}_\pm(h))$, then*

$$\|\mathcal{V}; L_2(\square_\pm^q)\|^2 \leq ch^2 \|\nabla \mathcal{V}; L_2(\square_\pm^q)\|^2, \quad (5.30)$$

$$\|\mathcal{W}; L_2(\square_\pm^q)\|^2 \leq ch \|\mathcal{W}; H^1(\mathcal{Q}_\pm(h))\|^2. \quad (5.31)$$

We apply the inequality (5.30) twice and the estimate (5.31) once to the terms in (5.29) containing h^{-2} , the inequality (5.30) and the estimate (5.31) to the terms in (5.29) containing h^{-1} , and the inequality (5.30) to the terms in (5.29) containing h^0 , we find

$$|\tilde{\mathcal{U}}^{q\pm}| \leq ch^{1/2} h^{1/2} \mathcal{N} \|\mathcal{R}; L_2(\square_\pm^q)\| \leq ch \|\mathcal{W}; H^1(\mathcal{Q}_\pm(h))\|. \quad (5.32)$$

The assumptions of Lemma 5.2 hold by the inclusion $w^\pm \in H^4(\Upsilon_\pm) \times H^3(\Upsilon_\pm)$ which implies the sufficient smoothness of terms of the asymptotic ansatz (cf. (5.20)). The extra factor $h^{1/2}$ in (5.32) is caused by integration over the transverse cross-section of the square $\square_\pm^q$. Since

$$\|\mathcal{R}; L_2(\square_\pm^q)\| \leq ch^{-1} \|\mathcal{R}; \mathcal{Q}_\pm(h)\|_\pm,$$

from the definition of the norm (5.5) it follows that

$$|\tilde{\mathcal{U}}_\pm^q| \leq ch^0 \mathcal{N} \|\mathcal{R}; \mathcal{Q}_\pm(h)\|_\pm \leq c \mathcal{N} \mathcal{E}(\mathcal{R}, \mathcal{R}; \mathcal{Q}(h))^{1/2}.$$

5.7. Theorem on asymptotics

In Subsections 5.4 and 5.5, we treated all the terms on the right-hand side of (5.32). The expression $c \mathcal{N} \mathcal{E}(\mathcal{R}; \Omega_h)^{1/2}$ is the common majorant in the obtained estimates. Thus, we have proved the following assertion.

Theorem 5.1. *If (5.14) and (5.15) hold, then the solution u to the problem (1.6), (2.2) is connected with the approximation $\mathbf{U}$, defined as the sum of the expressions (4.2) and (4.3), by the relation*

$$\mathcal{E}(u - \mathbf{U}; \Omega_h) \leq ch^0 \mathcal{N},$$

where the constant c is independent of $h \in (0, h_0]$ and f , whereas $\mathcal{N}$ is defined by (5.16).

By the Korn inequality (5.6), Theorem 5.1 yields an accuracy estimate for construction of the one-dimensional asymptotic model of junctions of beams with free external ends. By the standard scheme (cf., for example, [7, Section 5.4]), from the global asymptotic approximations (5.7) and (5.8) one can remove “extra” terms which were used for obtaining the system (2.28) (the lower order term $U^{1\pm}$ of the ansatz (2.6)) and the transmission conditions (the lower order conglomerate (1.11) and the boundary layer (3.15)). Furthermore, cut-off functions which are used for justifying the asymptotics are not necessary in final formulas. As a result, we also obtain, for example, the estimates

$$\begin{aligned} \|u_2^{h\pm} - h^{-2}w_2^\pm; L_2(\mathcal{Q}^\pm(h))\| &\leq ch^{1/2}\mathcal{N}, \\ \|u_1^{h\pm} - h^{-2}a_1^\pm - h^{-1}w_1^\pm; L_2(\mathcal{Q}^\pm(h))\| &\leq ch^{1/2}\mathcal{N}, \\ \|\sigma^\pm(u^{h\pm}) - A(D\mathcal{X}^\pm + \mathcal{Y}^\pm\mathcal{D}(\partial_z)w^\pm)w_2^\pm; L_2(\mathcal{Q}^\pm(h))\| &\leq ch^{1/2}\mathcal{N}. \end{aligned} \tag{5.33}$$

It is easy to verify that these estimates are asymptotically exact; moreover, they coincide with the estimates in [6], [7, Section 5.4] in the case of a single beam. In other words, the above modifications of asymptotic ansätze and methods for obtaining the transmission conditions in the one-dimensional model (which, of course, affect the solution to the limit problem) are displayed only in asymptotic analysis, but not in the theorem on asymptotics. The estimates (5.33) are also preserved in the case of boundary conditions on the external node $\mathcal{G}_h^\pm$ considered in Sections 2 and 4.

Acknowledgments

The work was supported by St. Petersburg State University (project No. 0.38.237.2014) and the Russian Foundation for Basic Research (project No. 15-01-02175).

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Submitted on September 5, 2015

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